

Tracer Test Report Stringfellow Acid Pits Superfund Site Jurupa Valley, California

**Contract Number: EP-S9-13-02
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**APTIM Project No. 500520
February 2023**

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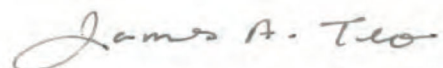
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Acronyms and Abbreviations

°C	degrees Celsius
°F	degrees Fahrenheit
μS/cm	microsiemens per centimeter
APTIM	Aptim Federal Services, LLC
a _y	transverse-horizontal dispersivity
a _z	transverse-vertical dispersivity
bgs	below ground surface
C	concentration (or electrical conductivity)
DTSC	California Department of Toxic Substances Control
EPA	U.S. Environmental Protection Agency
ft	foot (feet)
ft/day	foot (feet) per day
Kr	horizontal hydraulic conductivity
Krs	well skin conductivity
Kz	vertical hydraulic conductivity
GA	genetic algorithm
MCMC	Markov chain Monte Carlo
MNA	monitored natural attenuation
PCTF	Pyrite Canyon Treatment Facility
R ²	coefficient of determination
RAC	Remedial Action Contract
RWQCB	Regional Water Quality Control Board
ROD	Record of Decision
Site	Stringfellow Acid Pits Superfund Site, Jurupa Valley, California
Ss	specific storage
U	induced groundwater temperature change

1.0 INTRODUCTION

Aptim Federal Services, LLC (APTIM), under the Remedial Action Contract (RAC) III with U.S. Environmental Protection Agency (EPA), has prepared this report to document and evaluate investigative activities performed as specified in the work plan dated May 12, 2021 (APTIM 2021). Both the report and work plan were prepared according to the scope specified in RAC III Work Plan Addendum No. 2, Revision 1 for Remedial Investigation/Feasibility Study (RI/FS) Oversight dated April 29, 2020, and Work Variance Notification 002 (WVN002) dated April 19, 2021.

APTIM conducts oversight of the potentially responsible parties (PRPs), in this case the State of California represented by the California Department of Toxic Substances Control (DTSC), in completing the remedial investigation/feasibility study at the Stringfellow Acid Pits Superfund Site, Jurupa Valley, California (Site). The purpose of the oversight is to assist EPA in selecting a remedy that eliminates, reduces, or controls risks to human health and the environment.

EPA tasked APTIM to develop an analytical model for solute transport in a three-dimensional aquifer of finite thickness with uniform flow, dual porosity, sequential decay, and a finite-extent source with a specified time-variable concentration. The model also simulates heat transport in groundwater. The development of the mathematical model, coding in a programming language to provide a portable executable, testing, and preparation of a user manual are documented in a report dated May 2022 (APTIM 2022).

The objective of this work was to conduct a simple tracer test at the Site and demonstrate the applicability of the new analytical model for interpreting the observations from the tracer test.

1.1 Scope of Work

The scope of planned activities consisted of the following elements:

- Permits or permit exemption for injection of water with dissolved sodium chloride
- Preparation of injectant
- Tracer testing
- Test monitoring and data downloads
- Waste disposal
- Reporting

1.2 Disclaimer

The mentions of commercial equipment and/or software products or trade names throughout this report are for documentation purposes only and do not constitute endorsement or recommendation for use by EPA and/or APTIM.

1.3 Site Safety

Field activities were conducted in accordance with the Health and Safety Plan (Work Plan Appendix B, APTIM 2021).

1.4 Project Objectives

Based on the known site conditions and discussions with EPA, APTIM expects that the future site remedy for Zone 4 will include monitored natural attenuation (MNA) in addition to the active remedy. Characterization of contaminant transport by natural groundwater flow in Zone 4 will help evaluate the effectiveness of MNA. Therefore, wells in Zone 4 were selected for the tracer test. The test objective was to estimate aquifer properties related to solute transport in groundwater that can be used in future modeling for evaluation of MNA progress, predicting cleanup, etc. Tests were conducted at four wells as demonstration of the methodology, and the testing program may be later expanded to more wells in order to characterize contaminant transport in Zone 4 throughout the area of the Stringfellow plume.

The investigation objective was to conduct single well tracer tests on four wells and interpret the test results using the new analytical transport model. APTIM also used the results of historical and concurrent Site investigations and remediation activities to select the test wells and support the interpretation of the tracer testing. The overall scope and objectives were:

- Conduct single-well tracer tests on three selected wells by injecting heated water with dissolved sodium chloride, and on a fourth well by injecting heated distilled water.
- Record water level, temperature, and conductivity in the test wells for up to four weeks.
- Interpret the test results by fitting an analytical transport model and document all activities and results in a report.
- Dispose of wastes generated by field activities according to federal, state, and local regulations.

2.0 SITE CONDITIONS AND BACKGROUND INFORMATION

This section presents Site history and background information.

2.1 Site Location and Description

The Site is a former state-authorized Class I industrial waste disposal facility located within the city of Jurupa Valley in Riverside County, California. The location of the Site is shown on Figure 2-1. During the 16 years that it operated (from 1956 through 1972), approximately 34 million gallons of liquid industrial process waste were discharged into evaporation ponds at the facility. The industrial waste accepted at the facility comprised primarily of spent acids and caustics, metals, solvents, and pesticide by-products. Contamination in soil and groundwater by perchlorate resulted from historical testing activities in the Stringfellow Canyon and rock mining and testing in adjacent areas. The Site history and conceptual site model are described in detail in draft Stringfellow Technical Impracticability Evaluation (Ramboll 2020).

The Site has been divided geographically into four management zones (Figure 2-2), as follows:

- On-site/Upper Mid-Canyon Area (referred to as Zone 1), which includes the original 17-acre disposal area in the northern uppermost part of Pyrite Canyon, designated as Zone 1a, and an area extending approximately 600 feet (ft) southward of the clay barrier dam, designated as Zone 1b.
- Mid-Canyon Area (referred to as Zone 2), which is the portion of the Stringfellow plume located within the middle reach of Pyrite Canyon extending approximately 1,200 ft south of Zone 1. The mid-canyon extraction wells are located across the southern portion of Zone 2.
- Lower Canyon Area (referred to as Zone 3), which is the portion of the Stringfellow plume located within the lower reach of Pyrite Canyon extending approximately 2,000 ft south of Zone 2 to Highway 60. The lower canyon extraction wells are located along the southern, downgradient boundary of Zone 3.
- Community Area (referred to as Zone 4), which includes impacted groundwater migrating south of Highway 60 from Pyrite Canyon into the residential community of Jurupa Valley to just north of the Santa Ana River.

The lateral (east-west) boundaries of Zones 1, 2, and 3 are limited to the extent of contamination in soil and groundwater and generally correspond to the alluvial fill of the Pyrite Canyon. The extent of Zone 4 is wider, but the Stringfellow related contamination in groundwater is limited to a zone along the Pyrite Creek from Highway 50 in the north to 54th Street in the south.

In addition to these four zones, EPA has defined two additional areas in Pyrite Canyon as “EPA Area 1” and “EPA Area 2”. Investigations conducted between 2013 and 2017 in these areas indicated that additional sources of perchlorate are contributing to the groundwater plume originating from the Site.

To date, EPA has issued four interim Records of Decision (RODs) for the Site. A final ROD is in preparation and will be based in part on concurrent feasibility studies. The remedy for perchlorate and trichloroethylene (TCE) contamination in groundwater in Zone 4 is expected to rely in part on MNA.

2.2 Climate

The climate in the Stringfellow area is semiarid, typical of inland valleys of southwestern California. During the summer months, daytime high temperatures generally range from 90 degrees Fahrenheit (°F) to 100°F, with occasional highs greater than 100°F. Winter daytime high temperatures generally range from the 60s to the mid-70s, and winter nighttime lows below freezing are not uncommon.

2.3 Geology and Hydrogeology

The Site is underlain by three geologic units that are capable of storing and transmitting varying amounts of groundwater. These units include:

- Alluvium,
- Weathered bedrock, and
- Unweathered, fractured bedrock.

The alluvium consists predominantly of silts and sands, with interbedded layers of clayey sand and occasional layers of clean sand. It is a complex of debris flows, talus, alluvial fan deposits, and fluvial deposits. A paleochannel consisting mostly of sand and gravel trends subparallel to the present-day Pyrite Creek. In Zones 1 to 3, the alluvium thickness ranges from a few feet thick near the walls of Pyrite Canyon to approximately 114 ft thick in the deepest channels. Bedrock hills are also present in some areas in Zone 4.

The alluvium is underlain by bedrock consisting of granodiorite, quartz diorite, gabbro, and metasediments including quartzite, schist, and gneiss. The bedrock has weathered to varying degrees and depths across the Site area. The uppermost highly weathered to decomposed rock is referred to in this report as the “weathered bedrock.” The geologic characteristics of the weathered bedrock can vary greatly, with the size of lithic fragments increasing with depth from friable (easily crumbled) coarse-grained sand-sized particles to fist-size rocks. The thickness of the weathered bedrock varies depending on location, and it grades into unweathered bedrock at depth.

Groundwater flow occurs both as porous media flow through the alluvium and upper weathered bedrock units and as fracture flow through the deeper weathered bedrock and the unweathered bedrock units. Depth to groundwater in Zones 1 through 3 ranges from approximately 9 ft to over 130 ft below ground surface (bgs), and depth to groundwater in Zone 4 ranges from approximately 4 ft bgs to over 95 ft bgs.

The direction of groundwater flow within the alluvium and weathered bedrock units is generally to the south-southwest. Closer to the Santa Ana River, the groundwater flow direction shifts to the west, parallel

to the Santa Ana River, and then turns to the northwest towards large capacity production wells in the Chino Basin. The hydraulic gradient is highest in the uppermost area of the Pyrite Canyon and decreases in the downgradient direction. Groundwater in the bedrock unit is expected to generally follow a similar flow pattern as groundwater in the two overlying aquifer units.

The alluvium and weathered bedrock are considered to be hydraulically continuous. In general, the hydraulic conductivity of the alluvium is higher than that of the weathered bedrock, which in turn is higher than the bedrock. Hydraulic conductivity values from Zone 4 slug tests and pumping tests range from 1 foot per day (ft/d) to 130 ft/d with generally higher conductivity values for the alluvium than for weathered bedrock (Environ 2015).

2.4 Hydrology

The two major surface hydrologic features at the Site are Pyrite Creek and the Santa Ana River. Pyrite Creek is an ephemeral stream, originating in Pyrite Canyon in the Jurupa Mountains, and is the primary channel for surface water runoff in the study area. Pyrite Creek is mostly unlined and normally dry in Zone 4. During periods of high intensity or long-duration precipitation, the creek conveys runoff to its discharge point in the Santa Ana River. In addition to potential discharges from the surface water drainage channel that borders the original disposal facility and runoff from the canyon walls upstream of the formal disposal area, Pyrite Creek receives treated groundwater discharged from the Community Wellhead Treatment System (CWTS) in Zone 4.

The Santa Ana River, located at the southern limits of Zone 4, flows year-round. Under base flow conditions of approximately 45,000 gallons per minute (gpm), approximately 70 percent of the flow of the Santa Ana River is effluent from wastewater treatment plants.

2.5 Nature and Extent of Contamination

The original waste disposal area in Pyrite Canyon (Zone 1a) is the principal source of groundwater contamination at the Site. Additional sources of contamination include historical industrial activities on adjacent properties and agricultural activities in Zone 4. Specifically, past use of Chilean nitrate fertilizer resulted in soil and groundwater contamination throughout Zone 4. The contaminants and their distribution are summarized in detail in the draft Technical Impracticability Evaluation (Ramboll 2020). Recent investigations determined the extent of contiguous perchlorate plume from Zone 1 into Zone 4; the downgradient limit of this plume is near 54th Street (Kleinfelder 2020).

3.0 FIELD ACTIVITIES

Pre-field and field work activities are discussed in the following subsections. Field notes and photo log are included in Appendix A.

3.1 Permitting and Notification

APTIM notified DTSC about the schedule and nature of the anticipated work and coordinated scheduling of all field activities with DTSC representatives. Access to designated area for staging equipment and filling tank with water was only needed on four days. DTSC representative provided APTIM with a key to unlock wells.

In a May 10, 2021, letter to the Regional Water Quality Control Board (RWQCB) Riverside, California office, EPA satisfied the substantive permit requirements covering the introduction of salts into the Chino Basin by attaching Form 200 and used its authority under Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) to exempt the injection of water with dissolved sodium chloride from waste from paying the permit fees (EPA 2021).

Prior to initiation of field activities, APTIM notified DTSC, RWQCB Riverside office, and City of Jurupa Valley of the schedule and nature of the anticipated work.

3.2 Tracer Testing

Site preparation activities are discussed in the following subsections.

3.2.1 Reconnaissance

APTIM inspected the designated area behind the entrance gate to the Pyrite Canyon Treatment Facility (PCTF) for storing materials, source of municipal water on site, and visited the first two test wells, FC564A2 and FC951A1, in Zone 4 to inspect their condition on June 22, 2021.

APTIM visited FC1011A and FC1029A to inspect their condition on April 15, 2022.

3.2.2 Injection at Well FC564A2

APTIM placed 100 gallons of distilled water in five-gallon containers on a black plastic sheet laid on the ground in the designated area at the PCTF on June 28, 2021.

In the afternoon on June 29, 2021, APTIM transferred the water into a poly tank placed on the bed of APTIM pickup truck.

In the following discussion, the term injection is used for gravity draining water from the tank into the test well (i.e., the well was not pressurized). The injection of 100 gallons of distilled water in FC564A2 started at 4 pm on June 29, 2021, and lasted about 30 minutes, with water draining from the tank into the well.

Prior to water injection, APTIM used an Aqua TROLL 200 to read the electrical conductivity and temperature of the injected water in the tank, then lowered it into the well to record pre-injection conductivity and temperature of groundwater. While the probe equilibrated in the well, APTIM assembled the injection line, valve, and flowmeter. The Aqua TROLL 200 was programmed to record at 10-minute intervals. An additional transducer, an Aqua TROLL 700, was used also within well FC564A2 to record water level at high frequency (every 10 seconds) for hydraulic analysis; the probe also recorded temperature. The injection rate was recorded by taking time-stamped photos of the digital flowmeter. Leaves that were inadvertently left in the tank clogged the flowmeter and flowmeter readings were erratic. This affected the ancillary hydraulic analysis of the injection but not the tracer test. After the injection, the TROLL 700 probe was removed, and the well was closed with the Aqua TROLL 200 secured inside and continuing to record temperature and electrical conductivity.

The temperature of the injected water (37.8 degrees Celsius [$^{\circ}\text{C}$]) was 15.4 $^{\circ}\text{C}$ higher than the pre-injection temperature of groundwater (22.4 $^{\circ}\text{C}$). The pre-injection electrical conductivity of groundwater was 1,591 microsiemens per centimeter ($\mu\text{S}/\text{cm}$) higher than the conductivity of the injected water (calculated from recorded values after the test).

3.2.3 Injection at Well FC951A1

After flushing the tank, APTIM filled the 100-gallon tank with water from a faucet in the designated area at the PCTF on June 29, 2021, while adding a total of 8 pounds of food-grade sodium chloride. The water was left in the poly tank placed on the bed of APTIM pickup truck throughout the following day in full sun, including travel back to Stringfellow from Irvine.

Well FC951A1 was instrumented with an Aqua Troll 700 and an Aqua TROLL 200 programmed in the same way as for the test at FC564A2 and also the test procedures were the same.

Injection of 100 gallons of saline water into well FC951A1 started at 3:50 pm on June 30, 2021, and lasted approximately 30 minutes, with water draining from the tank into the well. The flowmeter was cleaned and checked before the test and functioned correctly. The flow rate varied over time and was frequently recorded by time-stamped photos. The valve was shut off for a few minutes half-way through the injection to introduce an additional variation into the injection rate for purposes of hydraulic analysis (this had no impact on the tracer test).

The temperature of the injected water (31.78 $^{\circ}\text{C}$) was 9.5 $^{\circ}\text{C}$ higher than the pre-injection temperature of groundwater (22.28 $^{\circ}\text{C}$). The pre-injection electrical conductivity of the injected water was 19,653 $\mu\text{S}/\text{cm}$ higher than the conductivity of groundwater (calculated from recorded values after the test).

3.2.4 Injection at Well FC1011A

After flushing the tank, APTIM filled the 100-gallon tank with water from a faucet in the designated area at the PCTF on August 4, 2022, while adding a total of 10 pounds of food-grade sodium chloride. The water was left in the poly tank placed on the bed of an APTIM pickup truck throughout the day in full sun.

Prior to water injection, APTIM used an Aqua TROLL 200 to read the electrical conductivity and temperature of the water in the tank, then lowered it into the well to record pre-injection conductivity and temperature of groundwater. While the probe equilibrated in the well, APTIM assembled the injection line, valve, and flowmeter. The Aqua TROLL 200 was programmed to record at 1-minute intervals. The injection rate was recorded by taking time-stamped photos of the digital flowmeter. After the injection, the well was closed with the Aqua TROLL 200 secured inside and continuing to record temperature and electrical conductivity.

Injection of 100 gallons of saline water into well FC1011A started at 3:36 pm on August 4, 2022, and lasted approximately 34 minutes, with water draining from the tank into the well. The flowmeter was cleaned and checked before the test and functioned correctly. The flow rate varied over time and was frequently recorded by time-stamped photos.

The temperature of the injected water (34.87°C) was 13.1°C higher than the pre-injection temperature of groundwater (21.79°C). The pre-injection electrical conductivity of the injected water was 30,000 $\mu\text{S}/\text{cm}$ higher than the conductivity of groundwater (calculated from recorded values after the test).

3.2.5 Injection at Well FC1029A

After flushing the tank, APTIM filled the 100-gallon tank with water from a faucet in the designated area at the PCTF on August 3, 2022, while adding a total of 10 pounds of food-grade sodium chloride. The water was left in the poly tank placed on the bed of an APTIM pickup truck throughout the day in full sun.

Well FC1029A was instrumented with an Aqua Troll 200 programmed in the same way as for the test at FC1011A and the test procedures were the same.

Injection of 100 gallons of saline water into well FC1029A started at 3:48 pm on August 3, 2022, and lasted approximately 47 minutes, with water draining from the tank into the well. The flowmeter was cleaned and checked before the test and functioned correctly. The flow rate varied over time and was frequently recorded by time-stamped photos.

The temperature of the injected water (35.67°C) was 13.9°C higher than the pre-injection temperature of groundwater (21.79°C). The pre-injection electrical conductivity of the injected water was 19,934 $\mu\text{S}/\text{cm}$ higher than the conductivity of groundwater (calculated from recorded values after the test).

3.3 Test Termination

APTIM visited test wells FC564A2 and FC951A1 on July 20, 2021, to download and inspect the recorded test responses. Because visual inspection of the downloaded electronic records clearly indicated that the groundwater electrical conductivity and temperature returned to pre-test values, APTIM terminated the tests, removed all equipment, and demobilized.

APTIM visited test wells FC1011A and FC1029A on August 15, 2022, to download and inspect the recorded test responses. Because visual inspection of the downloaded electronic records clearly indicated that the groundwater electrical conductivity and temperature returned to pre-test values, APTIM terminated the tests, removed all equipment, and demobilized.

3.4 Variances from the Work Plan

Due to scheduling, data loggers in wells FC564A2 and FC951A1 were not inspected weekly as planned, but three weeks after injections on July 20, 2021. Inspection of the data loggers revealed that the water quality (both electrical conductivity and temperature) at FC564A2 returned to pre-injection conditions on July 5, 2021, and at FC951A1 on July 9, 2021. The files download on July 20, 2021, included the recording of the entire test responses.

Due to scheduling and based on experience with the 2021 tests, the wells tested in 2022 were not inspected weekly, and the data loggers were retrieved almost one month after injections.

4.0 TEST EVALUATION

This section discusses the methodology of the test evaluation.

The main objective of the tests is to estimate aquifer properties that affect solute transport in groundwater and are important in assessing remedy performance (or selecting a remedy). These include Darcy flux, mobile and total porosity, dispersivity, and also the rate coefficient for mass transfer between mobile and immobile porosity. The interpretation also assesses which parameters are easier to resolve from the test observations to inform the planning of future tracer tests.

The recorded data are included in Appendix B.

4.1 Methodology

Each tracer test was conducted on a single well, i.e., the same well was used for injecting the tracer and for monitoring. The injected water with the tracer spreads into the aquifer around the well screen, creating an initial cylindrical zone containing the tracer around the well that disperses and migrates with groundwater flow away from the test well (Figure 4-1). Because the test does not rely on other observation wells and long travel distances for the tracer, its duration may be shorter than for multi-well tests. Because it relies on natural groundwater flow, it does not require pumping equipment and disposal of extracted water needed for push-pull tests.

Two tracers were used for each test: solute and heat. Specific electrical conductivity of water was used as proxy for solute mass concentration. Because of the small temperature change, conversion to temperature-corrected salinity was unnecessary. The change in solute concentration was achieved by injecting water with dissolved sodium chloride and by injecting distilled water. For each case, the introduced change in electrical conductivity was taken as the absolute value of the difference between pre-injection conductivity and conductivity of the injected water. The injected water was solar-heated above the temperature of groundwater, and the resulting temperature increase was monitored.

4.1.1 Conceptual Model of Tracer Injection

The injection is treated as a “plug flow” model where groundwater present in the portion of the pore space described as the effective porosity is displaced by the injected liquid. The effective porosity is a fraction of total porosity that allows movement of groundwater. The remaining fraction of porosity is filled with stagnant groundwater. For injection of a constant volume of water, the injected volume of aquifer is greater when the effective porosity is a small fraction of total porosity than when the effective porosity is a larger fraction of the same total porosity. Both the total and effective porosity of the aquifer are unknown. The tracer test is affected by bulk values of effective and total porosity of the undisturbed aquifer material, and these values may differ from their counterparts measured on soil samples in a laboratory that represent a much smaller volume and may be disturbed during their collection.

After injection into the saturated soil, the initial solute concentration in the effective (or mobile) porosity will be the same irrespective of the value of the effective porosity. Note: in this discussion, the terms “effective porosity” and “mobile porosity” are used interchangeably and represent the fraction of porosity that allows movement of groundwater; this fraction of total porosity may differ from drainable (called effective) porosity determined from geotechnical tests. The equilibrium temperature of groundwater (after thermal equilibrium between the injected water, groundwater in stagnant pores, and soil matrix is reached) will, however, depend on the volume of the injected aquifer zone, and the initial temperature increase will be smaller for lower effective porosity (and larger injected aquifer volume).

It is noted that near the edges of the injected zone the distribution of the tracer will be affected by dispersion. Plug-flow conditions are expected to prevail in the center of the injected zone around the well. The dispersion effect is expected to be minimal during the injection phase of the test, which was planned to be of short duration (and lasted about 30 minutes).

After the injection the injected zone is the initial concentration and temperature distribution, and the injected “plume” starts to migrate with groundwater flow. As the plume moves downgradient, the concentration and temperature at the injection well will gradually decrease. The migration will be affected by groundwater velocity, total and effective porosity, mass transfer rate between the mobile and immobile porosity fractions, and dispersion (including molecular diffusion). The chosen tracer (sodium chloride) is non-reactive so that first order reactions will not affect its transport.

4.1.2 Well Construction

Multiple monitoring wells and piezometers exist in Zone 4 (Figure 4-2). Wells FC564A2 (Figure 4-3), FC951A1 (Figure 4-4), FC1011A (Figure 4-5), and FC1029A (Figure 4-6) have been selected based on their construction, depth to groundwater, electrical conductivity of groundwater (Kleinfelder 2019), and hydraulic conductivity of the aquifer at their location (Environ 2015). All four wells are screened in the alluvium. The depth to water was measured on the injection days (June 29 and 30, 2021, and August 3 and 4, 2022):

Well Name	Aquifer Thickness (ft)	Screen Depth (ft bgs)	Depth to Water (ft)	Nominal Casing Diameter (in)	Inside Casing Diameter (in)	K (ft/d)	Groundwater Electrical Conductivity (mS/cm)	Tracer
FC564A2	21.12	20.4–61.6	43.88	4.5	4.026	82	1.5	Distilled water
FC951A1	27.3	23–28	21.01	6	5.761	57	1.2	Sodium chloride
FC1011A	47.66	15–60	17.47	4	3.75	40	1.5	Sodium chloride
FC1029A	35	53–83	17.32	6	5.67	13	1.5	Sodium chloride

Notes:

ft – feet

ft/day – feet per day

ft bgs – feet below ground surface

mS/cm – milliSiemens per centimeter

in – inches

4.1.3 Predictive Simulations

Predictive simulations were used to estimate the necessary volume of injectant, duration of the tracer test, and the sensitivity of the test response to key transport parameters to help plan the test procedures (APTIM 2021). The simulations were conducted for at FC951A1. Using the reported aquifer and well properties and assumed transport parameters, the necessary volume of the injectant that would cause sufficient initial temperature/conductivity change without necessitating very long test duration was estimated to be about 100 gallons. Either distilled water or water with dissolved sodium chloride would provide sufficient electrical conductivity contrast to act as tracer. For heat transport, the injected water should be at least 10°C warmer than the groundwater to result in a sufficient temperature contrast; the groundwater is at about 20°C. The solute and heat tracer test was expected to last one week before the concentrations and temperature in the well return to pre-test conditions. Because of uncertainty in the aquifer properties and transport parameters, the migration of the tracer may be faster or slower, and the test duration could extend to several weeks.

The simulations also show the sensitivity to several key parameters related to solute and heat transport, such as porosity and effective porosity, Darcy flux, dispersivity, and mass exchange rate between mobile and immobile porosity. The simulated test responses display different time-concentration or time-temperature curves for changed values of the transport parameters; this is an indication that the transport parameters can be resolved by fitting the model to the test observations. Insensitive parameters would result in similar simulated responses, indicating that they cannot be resolved by model fitting. The combination of solute and heat tracers will enhance the sensitivity of the test to Darcy flux and total and effective porosity.

The tracer test observations were simulated using an analytical transport model for a volumetric source in a three-dimensional aquifer of finite thickness with dual porosity referred to as Model 2 (Perina 2022; APTIM 2022) and its simpler version for a single porosity aquifer referred to as Model 4. Both models can simulate either solute or heat transport in groundwater.

4.1.4 Ancillary Evaluation of Injection as Aquifer Test

The injection offers an ancillary opportunity to treat it as a hydraulic test for estimation of aquifer properties, mainly hydraulic conductivity. The water level response to injection is the reverse to drawdown response during pumping, and the test can be treated as a variable rate pumping test. One important difference in well performance during extraction and injection is that frictional losses across most well screens are higher during injection than during extraction because of the shape of screen openings; this difference is significant for high rate injection and extraction, but for low flow rates frictional losses are negligible. The frictional well loss coefficient can be estimated as an additional parameter from time-drawdown data.

The results from this analysis, specifically the hydraulic conductivity in the radial (horizontal) direction, in combination with measured groundwater gradient, will serve as an additional, independent check on the horizontal Darcy flux in the aquifer.

Monitoring of water level in the test well requires high frequency recording with a pressure transducer/data logger, and monitoring of the time-varying injection rate requires a (digital) flowmeter on the discharge hose or pipe. The flow rate must be controlled by a valve to avoid too rapid draining of the tank.

4.2 Data Pre-Processing

The data-logger records were imported into a spreadsheet to inspect and prepare observation input files for inverse modeling.

4.2.1 Water Level Data Pre-Processing

The recorded time series of water levels were visually inspected, corrected for starting time, and referenced to initial water level to obtain “time-drawdown” data sets. The data recorded at high frequency were interpolated and sub-sampled with a smoothing filter at a reduced number of points. The time-flowrate data pairs were used as recorded (without reduction).

The flow totalizer readings were tabulated, converted to flow rate, and time scale corrected for starting time to match the water level observations time series. Manual adjustments of the flow rate observations were necessary for the injection to FC564A2 because the flowmeter readings were disrupted; these adjustments were constrained by the total injected volume so that the sum of multiples of time interval and flow rate equals 100 gallons.

The injection rate and time-drawdown observations for FC564A2, FC951A1, FC1011A, and FC1029A are shown on Figures 4-7 through 4-10.

4.2.2 Tracer Test Data Pre-Processing

The recorded electrical conductivity (internally corrected for temperature by the probe used) and temperature data were visually inspected and corrected for starting time.

The conductivity was referenced to pre-injection conductivity in the well. The conductivity “signal” used as a proxy for concentration is the absolute value of the difference between pre-injection conductivity in the well and the conductivity of the injectant.

The temperature data set was referenced to the difference between the temperature of the injected water and the pre-injection temperature of groundwater. The temperature “signal” is the difference between the initial temperature and pre-injection temperature of groundwater.

The referenced conductivity and temperature observations were interpolated and sub-sampled with a smoothing filter at a reduced number of points; 10 and 50 points were used. This step is necessary to limit the computational effort in fitting the transport model to observations: the model must compute a simulated value for each observed value, so reducing the number of observations reduces the execution

time. It is noted that the “information content” of the entire data set is generally preserved in the reduced data set.

The original and subsampled time-conductivity and time-temperature observations for FC564A2, FC951A1, FC1011A (without subsampled data), and FC1029A are shown on Figures 4-11 through 4-14.

4.3 Hydraulic Analysis

The water level rise in the test well due to injection was treated as negative drawdown and the injection analyzed as a variable rate pumping test. The time-drawdown observations from the injection and recovery phase were fitted using an analytical model for flow near a pumping well that accounts for partial penetration, wellbore storage, frictional well loss, aquifer anisotropy, and variable flowrate calculated by superposition of linear segments (Perina and Lee 2006; Perina 2020). The estimated aquifer properties include horizontal (in the radial direction, K_r) and vertical hydraulic conductivity (K_z), and specific storage (S_s). Rather than estimating K_z , the ratio K_z/K_r was estimated instead because it can be constrained to be smaller than 1. Because of the short duration of the injection, the aquifer response is identical to the response of a confined aquifer; unconfined aquifer response could be simulated but was not needed.

The main result of the hydraulic test is the horizontal (radial) hydraulic conductivity. The conductivity and horizontal groundwater gradient from the last (2020) groundwater monitoring event is used to calculate the value of Darcy flux independently of the tracer test results. This comparison serves as validation of the tracer test results.

The remaining estimated aquifer properties (S_s and K_z/K_r) in general cannot be reliably determined from single-well tests and their values are uncertain.

4.4 Transport Analysis

Multiple inverse simulations were conducted to test what parameters may be reliably estimated from the data set using the transport model and to recommend model and data combination for practical use.

Inverse modeling involves modifying model inputs to achieve an acceptable fit between observations and their model-computed values. The inverse modeling (or fitting) used Model 4 for mass and heat transport, and Model 2 for mass transport accounting for dual porosity. Model 2 accounts for dual porosity with first order mass exchange between the mobile and immobile porosity and a source that may be time-dependent. Model 4 is for an instantaneous release of mass or heat in an aquifer of single porosity. Heat transport (with the assumption of local temperature equilibrium between water and solids) is adequately simulated with Model 4. Solute transport is more accurately simulated with Model 2. Models 2 and 4 are for an aquifer of finite thickness and width, and infinite extent along the direction of groundwater flow. Model 4.1 is a modification of Model 4 and represents an infinitely wide aquifer.

Fitting of respective observations with heat and mass transport models was done both separately and simultaneously. Separate fitting of heat and mass transport models was used to quantify differences in estimated parameters for a scenario when only one type of tracer test is conducted.

The fitting of a model to observations is a solution to an inverse problem that can be accomplished using various methods to determine the values of the (multiple) unknown model parameters. For this work, the inversion used a sequence consisting of genetic algorithm (GA) followed by Markov chain Monte Carlo (MCMC). The GA is a mathematical representation of evolution of species; it starts with a randomly generated population consisting of individual sets of the unknown parameters and proceeds with selection (better solutions have a higher chance of survival), mutation (random change), and cross-over (combination and inheritance of parent properties) to generate the next population. The GA searches a wide range of possible parameter values and the user does not need to supply initial guesses. Each successive population contains better solutions to the inverse problem. The best solution from the GA is passed on as a starting point for the MCMC. This “best solution” does not need to achieve a close fit to observations, just a fit good enough to start the MCMC. The MCMC steps from one solution to the inverse problem (i.e., one set of parameters) to another, randomly generated solution with a probability that depends on the goodness of fit between the model and observations. The chain converges on the distribution of the parameters that solve the inverse problem and allows estimation of their standard deviations and correlations. Perina (2020) demonstrated the effectiveness of the GA-MCMC sequence in analysis of aquifer tests. The code for the GA and MCMC is from GWFi (Copyright 2020), a program for inverse analysis of aquifer tests. All parameters were estimated in log-space, i.e., the GA and MCMC operated on the natural logarithms of the parameter values. The reported results are means from the last of three sequential chains of length 1,000.

The drawback of this GA-MCMC approach is the requirement to evaluate the model many times. This is not an issue for Model 4 that is time-domain and its execution is fast, but Model 2 includes numerical Laplace transform and its execution is about two orders of magnitude slower. For example, an inverse simulation using Model 4 that included over 13,000 forward simulations took about 2–3 minutes, while inverse simulation using Model 2 for the same data set and number of model executions required about 20 hours (one forward simulation in this context is the model calculation of output values for all observation data points). For this evaluation, the GA used population size of 99 and 10+1 generations, and the MCMC used sequential chains of total length 2,100 or 3,100.

The unknown aquifer properties estimated by inverse modeling of the observations included the Darcy flux, porosity, fraction of mobile porosity, first order rate constant for exchange between the mobile and immobile porosity, and dispersivity.

Some model parameters are not independent of each other and cannot be independently estimated by inverse modeling. A typical example is dispersivity that is usually defined as longitudinal (in the groundwater flow direction), transverse-horizontal (perpendicular to the flow direction), and transverse-vertical. The transverse dispersivities are smaller in magnitude than the longitudinal dispersivity, with

transverse-vertical dispersivity (a_z) being smaller than transverse-horizontal dispersivity (a_y). Such unknown model inputs may be treated as tied parameters and their value is calculated from the value of another estimated parameter. In this work, a_y and a_z were specified as 0.1 and 0.01 multiples of the longitudinal dispersivity (a_x) and continuously adjusted during the inversion process.

The shape of the injected zone was approximated by a rectangular prism with a height corresponding to the screen interval of the injected well and volume equivalent to the injected aquifer volume. The size of this initial concentration (or temperature) zone was recalculated with each change of porosity during the inversion process. Sensitivity simulations (APTIM 2021) showed that although the actual shape of the injected zone is cylindrical, the response at the injection well is the same for both a prism and cylinder shaped initial concentration distribution. This would not necessarily be the case if an observation well at some distance from the injected well was used.

Recording the test response at the injection well itself requires accounting for the volume of the water inside the well casing to accurately calculate the heat transport result. Water inside the well casing is initially at the temperature of the injected water while the temperature of the water injected into the aquifer decreases (almost instantly) due to heat exchange with the aquifer matrix and groundwater in immobile pores. The response inside the injected well is calculated by superposition of two sources—one is the injected zone in the aquifer with equilibrium initial temperature, the other is the wellbore with initial temperature equal to the temperature of the injected water minus the aquifer initial equilibrium temperature. The size of the injected zone and its initial equilibrium temperature are recalculated with each change in porosity during the inversion process. The wellbore (and screen and sand pack) have different properties than the aquifer and neglecting this difference introduces a small error in the calculated temperature.

The predominant minerals in the soils are quartz and feldspars. The thermal conductivity and specific heat capacity values, found in literature, are similar for both minerals. Consequently, the thermal properties of the aquifer material were treated as known. The corresponding thermal properties of water are also known. This simplifies the heat transport model because the only unknown parameters are the Darcy flux, porosity, effective porosity, and dispersivity.

For convenience of using published thermal properties of materials, the transport model input variables and observations were expressed in SI units. The estimated Darcy flux and other parameters were then converted to ft/d, etc.

5.0 RESULTS AND DISCUSSION

This section presents the results of analyses described in Section 4.

5.1 Hydraulic Analysis Results

The well construction with depths referenced to the top of the aquifer (water table) is shown in Table 5-1, and the estimated aquifer properties are shown in Table 5-2. Examples of detailed analysis results are shown in Appendix C.

The model fit to the observations for FC564A2 injection (Figure 5-1) is affected by data noise with Pearson's coefficient of determination (R^2) of 0.978. The estimated K_r for well FC951A1 is 204 to 211 ft/d which is higher than the values reported from slug tests and pumping tests, 71 ft/d and 80 ft/d, respectively (Environ 2015). The estimated K_z/K_r of 2.8×10^{-2} to 1.0×10^{-1} and S_s of 2.8×10^{-6} to 3.4×10^{-6} ft⁻¹ are plausible for the alluvium in Zone 4. An additional simulation used aquifer thickness equal to the combined saturated thickness of alluvium and weathered bedrock (instead of the saturated thickness of alluvium) and resulted in a K_r value of 172 ft/d, still higher than the historical test results.

The model fits the observations for FC951A1 injection very closely (Figure 5-2) with Pearson's R^2 of 0.996. The estimated K_r for well FC951A1 is 50 ft/d which is very close to the values reported from slug tests and pumping tests, 45 ft/d and 57 ft/d, respectively (Environ 2015). The estimated K_z/K_r of 1.6×10^{-2} and S_s of 1.1×10^{-4} ft⁻¹ are plausible for the alluvium in Zone 4 although the S_s value is at the high end of expected range.

The model fitting for FC1011A is rather poor (Pearson's R^2 of 0.979) indicating the actual aquifer conditions are different from the conceptual model (i.e., uniform aquifer). The well is installed in a heterogeneous aquifer with a gravel layer at the upper part. The estimated K_r for well FC1011A is 45 ft/d, which is in good agreement with the value reported from slug test results of 40 ft/d (Environ 2015). The estimated value S_s of 1.2×10^{-3} is too high. The estimated value well skin conductivity K_{rs} of 202 ft/d is high, suggesting an overdeveloped zone.

The FC1029A model fitting shows minor systematic misfit that is likely due to inaccuracy of flowrate measurement. The overall fit to observations is good (Pearson's R^2 of 0.991). The estimated K_r of 12 ft/d is higher than K_r value reported from slug tests of 3.3 ft/d (Environ 2015). The estimated K_{rs} of 2.3 ft/d is lower than K_r . This suggests the slug tests were affected by low permeability skin. The estimated value S_s of 4.6×10^{-6} is physically plausible. Also, the estimated K_z/K_r ratio of 1.2×10^{-3} is plausible, and its low value may be due to the presence of clay and silt lenses. The screen is within sands and silty sands with clay and the well was installed with a hollow stem auger rig, so some smearing of clay along the wellbore may have accounted for the low conductivity skin. Recovery was faster than at FC1011A, while K_r is lower.

5.2 Transport Analysis Results

The transport analysis results are summarized in Table 5-3. The model fit to observations is shown on Figures 5-3 and 5-4. Very similar goodness of fit was achieved for all simulations, therefore, figures are not shown for individual simulations. Examples of detailed analysis results are shown in Appendix D.

For some simulations, the total porosity was estimated (i.e., unknown parameter), and for others it was treated as known model input. This is because the total porosity is limited to a rather narrow range of plausible values for sandy soils (generally 0.3 to 0.4) unlike the effective (or mobile) porosity that can be a very small or quite large fraction of the total porosity, depending on soil grain distribution, soil matrix structure, roundness, etc. When total porosity was set to a specified value, it was 0.3, 0.35, and 0.4. When total porosity was estimated as an unknown parameter, simulations that resulted in implausibly low total porosity (lower than 0.2) were considered non-representative.

Darcy flux, fraction of mobile porosity, and longitudinal dispersivity were estimated in all simulations. Simulations using Model 2 for mass transport also estimated the mass transfer rate between the mobile and immobile porosity. Transverse dispersivities (α_y and α_z) were treated as tied parameters as described in Section 4.

Secondary transport parameters were calculated in Table 5-3 from the estimated parameters. Mobile porosity was calculated as the multiple of mobile porosity fraction and total porosity. Advective velocity was calculated as the ratio of Darcy flux to the mobile porosity.

5.2.1 Results for FC564A2

The results for FC564A2 were generally similar for a wide range of simulations. More consistent results were obtained for combined inversion of mass and heat transport with Models 2 and 4, respectively, than for individual simulations of either mass or heat transport, or for combined inversion with Model 4 used for both mass and heat transport.

Somewhat unexpectedly, the values for the Darcy flux were very similar for all simulations (about 0.8 ft/d), including individual inversion of mass and heat transport. Heat and mass tracer tests usually lead to different Darcy flux estimates.

The estimated value of total porosity (0.4) is plausible. The advective velocity is between 3.0 and 3.4 ft/d. The mobile porosity is about 0.2 to 0.3 and longitudinal dispersivity 0.03 to 0.18 ft. The mass transfer rate between the mobile and immobile porosity is 0.44 to 1.5 d⁻¹.

5.2.2 Results for FC951A1

The results for FC951A1 were more varied across the range of simulations than the results for FC564A2. More consistent results were obtained for combined inversion of mass and heat transport with Models 2 and 4, respectively, than for individual simulations of either mass or heat transport, or for combined inversion with Model 4 used for both mass and heat transport.

The estimated values of total porosity are much lower than would be expected for sandy soils and are not considered representative; total porosity could not be determined from the inversion of this test. The values for the Darcy flux estimated from heat transport only were unreasonably high. The results from combined inversion of mass and heat transport with Models 2 and 4, respectively, are considered representative.

The Darcy flux is about 0.3 ft/d. The advective velocity is between 1.2 and 1.4 ft/d. The mobile porosity is about 0.19 to 0.26 and longitudinal dispersivity 0.07 to 0.12 ft. The mass transfer rate between the mobile and immobile porosity is 0.29 to 0.49 d⁻¹.

5.2.3 Results for FC1011A

The recorded increases in electrical conductivity C (used as proxy for concentration) and temperature change U are much smaller than expected for injecting 100 gallons of water; predicted values based on mixing calculation are U = 7.5–9.7°C and C = 18,600–24,100 µS/cm while recorded values are U = 0.47°C and C = 1,900 µS/cm.

The recorded response shown on Figure 4-13 suggests that the injected water entered the aquifer above the TROLL probe and only limited mixing in the wellbore occurred below this entry zone. Depth to water was 17.4 ft. The TROLL was at about 40 ft. There is a zone of coarse arkosic sand with gravel at 32–37 ft and gravel was also noted at 20 ft and 27 ft. The lithologic log shows medium to coarse arkosic sand from 12–32 ft, sand and gravel 32–37 ft, sand 37–43 ft, gravel 43–45 ft, and clayey sand and silt 45–65 ft (bottom of alluvium). The zone below 45 ft is clearly less permeable.

Due to the injected water moving out of the well via a high hydraulic conductivity (K) zone above the TROLL, this test was not analyzed with a transport model.

5.2.4 Results for FC1029A

The results for FC1029A varied across the range of simulations. Simultaneous fitting of conductivity and temperature observations with mass and heat transport models, respectively, provided more physically reasonable results than individually fitting mass or heat observations only. Heat transport only clearly overestimates q; simulation of mass transport only likely underestimates phi and q.

Pearson correlation coefficients between computed and observed values are between 0.95 and 0.999. For simultaneous fitting of U and C, there is a systematic misfit for U (and R² = 0.95 for U while R² = 0.99 for C). Fitting for U only results in good fit (R² = 0.99). Correlations between estimated parameters are generally low.

The Darcy flux is about 0.24 ft/d. The advective velocity is about 3.5 ft/d. The mobile porosity is about 0.070 and longitudinal dispersivity 0.91 ft. The mass transfer rate between the mobile and immobile porosity is 0.02 d⁻¹. The value of total porosity (0.35) was not estimated but assumed.

Mixing calculations used a model for transient mixing of water in wellbore with injected water, with concentration (or temperature) in the well gradually increasing during injection, and a simple mass-volume (heat-volume) balance. The results suggest that water injected by gravity draining into the well mixed rapidly with water in the wellbore and sand pack. Calculated C matches C (10,600 $\mu\text{S}/\text{cm}$) increase measured in the well when using simple mass and volume balance, while a mixing model based on gradual instantaneous mixing in wellbore overpredicts the C at the end of injection by 3,000 $\mu\text{S}/\text{cm}$. The computed U was higher than recorded U (4°C) by 3.4°C and 5.4°C for the simple balance and gradual mixing models, respectively. The results suggest loss of heat from the injected water to the well casing during the draining phase when water trickled along the casing. The recorded U and C kept increasing after the end of injection; this may be indicative of eddy dissipation and mixing within the wellbore.

5.3 Comparison of Hydraulic Analysis and Transport Analysis Results

The results from the hydraulic analysis were used to calculate Darcy flux for each well location. The hydraulic gradient in the alluvium in the test area does not generally change year to year and was 0.0080 in 2019. The comparison is shown in Table 5-4.

For FC564A2, the Darcy flux values from the tracer test and historical hydraulic tests are 0.8 ft and 0.64 ft/d, respectively. These values are very close, likely well within the range of uncertainty. The Darcy flux value of 1.6 ft/d calculated from the injection test results appears to be high. However, it would correspond to advective velocity of over 6 ft/d, which is within the range of estimated advective velocities for this portion of Zone 4 (Kleinfelder 2011).

For FC951A1, the Darcy flux values from the tracer test and hydraulic test are 0.3 ft/d and 0.4 ft/d, respectively and 0.46 ft/d from historical hydraulic tests. These values are very close, likely well within the range of uncertainty.

For FC1029A, the Darcy flux values from the tracer test and hydraulic tests are 0.24 ft/d and 0.20 ft/d, respectively. These values are very close, likely well within the range of uncertainty.

The Zone 4 Remedial Investigation report, Section 4.2.4 (Kleinfelder 2011) includes estimated advective velocities and plume migration rate. Advective velocities were calculated from hydraulic conductivity values estimated from slug tests, groundwater gradients determined from water levels, and assumed values of effective porosity. For the northern part of Zone 4 (north of 54th Street), calculated advective velocities ranged from 0.040 ft/d to 7.40 ft/d. Darcy flux values, calculated from the presented values of advective velocity and effective porosity range from 0.012 ft/d to 1.5 ft/d. The tracer test derived advective velocity and Darcy flux values are within this range.

5.4 Comparison of Transport Analysis Results and Plume Migration

Kleinfelder (2011, Section 4.2.4) calculated an apparent plume migration rate based on the 2006 plume's length. Based on the length of the plume measured between the southern boundary of Zone 1 to the

Santa Ana River (approximately 25,000 ft) and assuming that perchlorate migration in groundwater started in 1956 (time interval of 50 years), the migration rate of perchlorate is 1.4 ft/d. This apparent plume migration rate is expected to be greater than advective velocity because the calculated migration rate includes the effect of dispersion. If perchlorate migration in groundwater started later, the rate would be higher. If the plume reached a steady state prior to 2006, it would also mean the migration rate was higher. On the other hand, this migration rate is “average” over the plume length and the actual rate may be lower or higher at different locations within the plume.

The apparent plume migration rate of 1.4 ft/d is nearly the same as the advective velocity estimated from the FC951A1 tracer test (1.2 to 1.4 ft/d), while the advective velocity values derived from the FC564A2 and FC1029A tracer tests are approximately twice as high. The velocities determined from the tracer tests represent local values, while the approximate, plume-derived migration rate represents large scale plume movement. The results are in agreement.

6.0 CONCLUSIONS

The test analysis demonstrated that a single-well tracer injection can be used to estimate key aquifer properties related to solute transport. Simultaneous use of mass and heat tracers followed by simultaneous analysis of mass and heat transport leads to improvement of the estimated aquifer properties when compared to using a single tracer (mass or heat).

Injection of a small volume of water, monitoring in a single well, no need for disposal of extracted water, and short test duration allow for a low cost tracer test. Because the test requires very limited equipment and can be completed in a relatively short time, it offers an alternative to natural gradient multi-well or forced-flow push-pull or cross-well tracer tests.

The injectant migrated away from each well in about one week. Therefore, the injections had no effect on annual groundwater monitoring and would not have affected even higher frequency (such as quarterly) monitoring program.

Because the injected well is monitored, the test will always record the tracer. In comparison, multi-well natural gradient tests may be affected by local heterogeneities, and the tracer may not be adequately intercepted at observation wells or may not be intercepted within an expected time frame.

The single well tests are of course affected by aquifer heterogeneities that could skew the results. For this reason, wells with relatively short screens are preferred to minimize in-well vertical flow and effects of heterogeneities in the vertical direction, such as in layered aquifers (an illustrative example is the test on FC1011A).

The estimated Darcy flux and advective velocity values are in good agreement with values derived independently during this investigation and values reported historically for Stringfellow Zone 4.

The estimated values for porosity, mobile porosity, and mass transfer coefficient between the mobile and immobile porosity are physically plausible for the sediments in Zone 4 alluvium. The estimated dispersivity values are reasonable for the small scale of the tracer migration and the fact that the model accounted for dual porosity.

The estimated values for total porosity were physically plausible, with realistic values for the test on FC564A2 (about 0.4) and too low values for the tests on FC951A1 and FC1029A (0.10 and 0.19, respectively). The tests did not allow for reliable determination of total porosity. Future work should focus on improving the test design to allow more reliable determination of total porosity. Among the possibilities to be considered are repeating the test with a different volume of injected water or using a sorbing tracer in addition to a non-sorbing tracer and heat. Sensitivity simulations would help assess these test modifications and identify conditions that improve parameter resolution.

The estimated values of dispersivity are low when compared to values used in a site-wide numerical solute transport model. This is partly due to the small scale of transport during the test but also because the test analysis accounted for dual porosity (dual porosity has a similar effect as dispersivity), while the numerical model used single porosity.

Analysis of the tracer test using an analytical (or semi-analytical) transport model is convenient because such model does not require discretization (and associated concerns about numerical accuracy).

The evaluation of tracer injection as an aquifer hydraulic test demonstrated that a relatively short-term and low volume injection by gravity-draining a tank provides a viable alternative to other aquifer tests; injection of 100 gallons over 30 minutes generated adequate water level response in the test wells. The injection also avoids the need for disposal of contaminated water extracted during pumping tests. The test requires frequent or continuous flow monitoring. The estimated aquifer properties were comparable to pumping test results. Due to low injection rates, frictional well loss is negligible. The fitting model must account for wellbore storage and variable injection rate.

It is preferable to conduct tests on wells at which aquifer tests were performed so that independent estimates of Darcy flux can be compared to the tracer test results.

The hose used to drain the injected water from the tank did not reach all the way to the water level in the well, and as water trickled along the well casing there was heat exchange with the casing; due to this heat loss, the initial temperature in the well was lower. The temperature in the well was still sufficiently higher than the pre-test aquifer temperature to allow for test interpretation. To avoid heat loss to the casing and get maximum temperature change in the aquifer, the drain hose (or pipe) should extend to the water table and should be of larger diameter to allow higher flow rate.

It may be needed to conduct this type of tracer test on wells that have a long water column above the top of the screen. The test analysis then must account for the mixing of the injected water with the water in the wellbore and for the mass and heat exchange with the water in the casing above the screen after the tracer injection. This condition is currently not accounted for in the model. Alternatively, the mixing may be minimized by instrumenting the injection piping with a seal against the casing.

7.0 RECOMMENDATIONS

It is desirable to improve the estimation of total porosity and mass exchange rate coefficient from the tests. Sensitivity simulations should be prepared to identify test design that increases the sensitivity of the simulated concentration and temperature to the total porosity. The test design changes may include adding a sorbing tracer and repeating the test with a different volume of injected water.

It would also be beneficial to assess the results of this single-well tracer test independently of the observations from the injected well. One possibility is to utilize an observation well that is located within the injected zone; the concentration and temperature response in such observation well would be more sensitive to porosity than the response in the injected well. If available, a well pair should be selected where the observation well is a short distance downgradient of the injected well. The local direction of groundwater gradient would, however, present an additional uncertainty in the test evaluation. The injected volume may also need to be increased to ensure the injected zone extends to the observation well.

A methodology for conducting and evaluating the single-well natural gradient tracer test at wells with deeply submerged screens should be developed.

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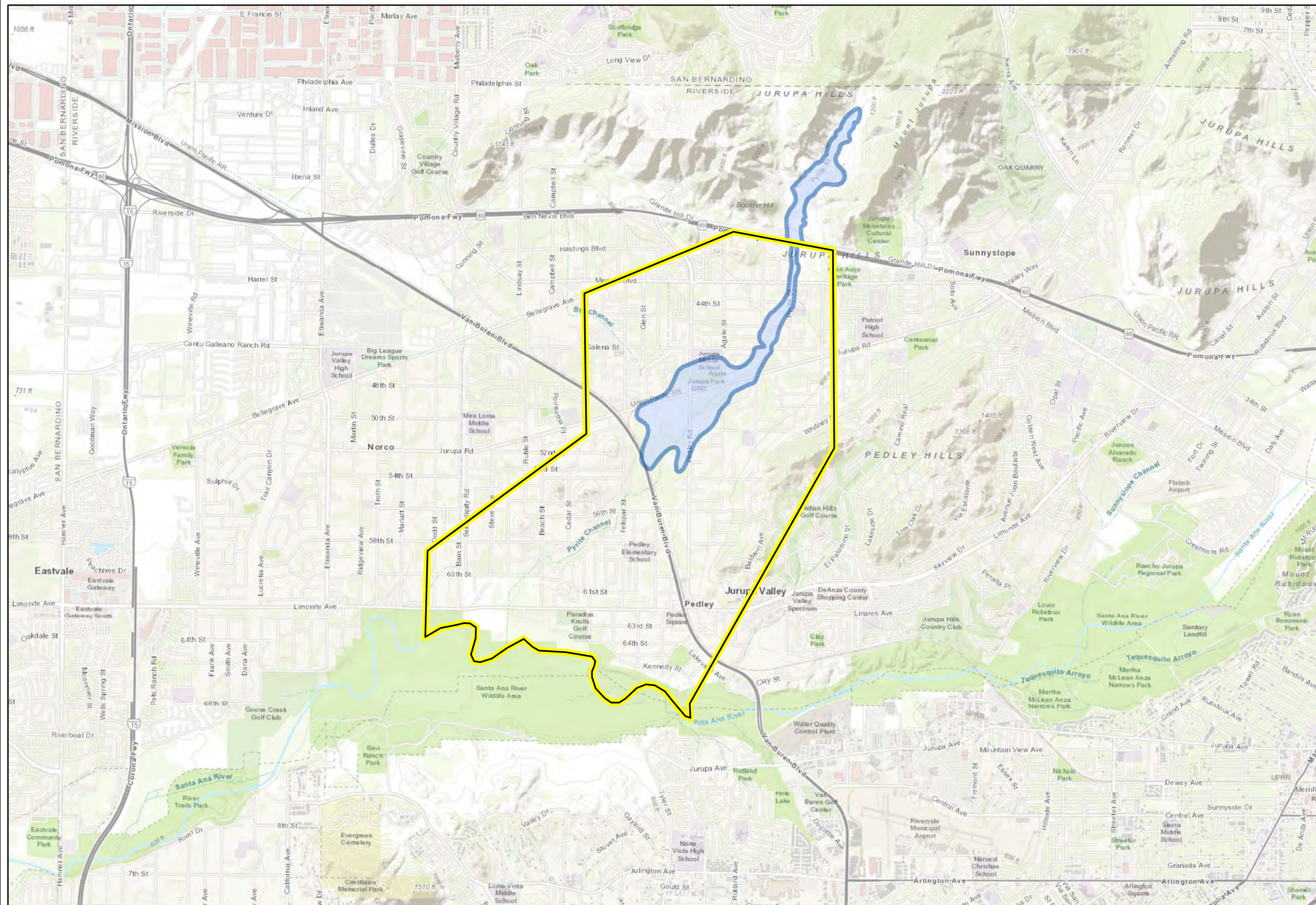
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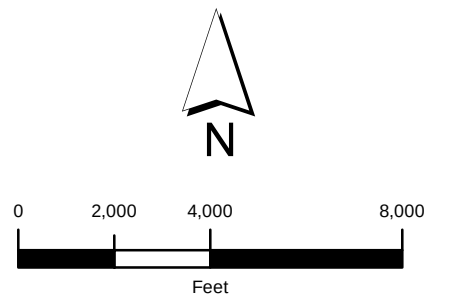
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Figures



LEGEND

- Zone 4
- Distribution of Pyrite Canyon Plume Exceeding Clean-up Levels



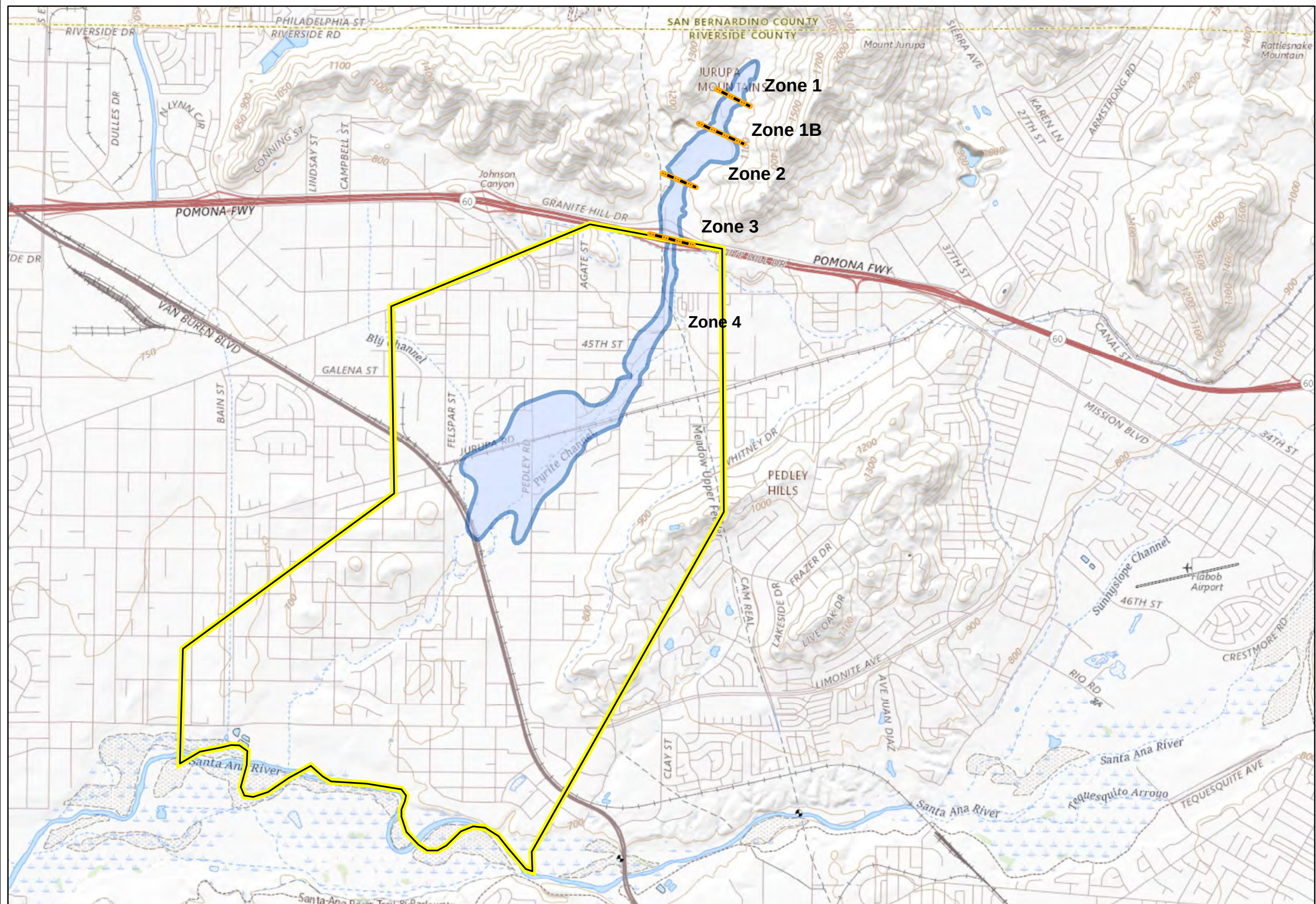
1 inch = 4,000 feet



Site Location Map

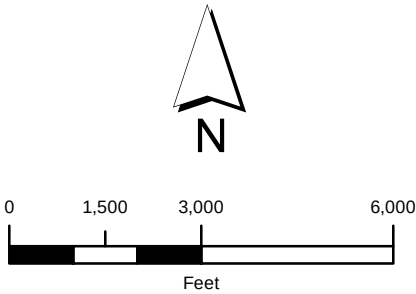
Stringfellow Superfund Site
Jurupa Valley, California

Figure
2-1



LEGEND

- Delineation of zone extent
- Zone 4
- Distribution of Pyrite Canyon Plume Exceeding Clean-up Levels

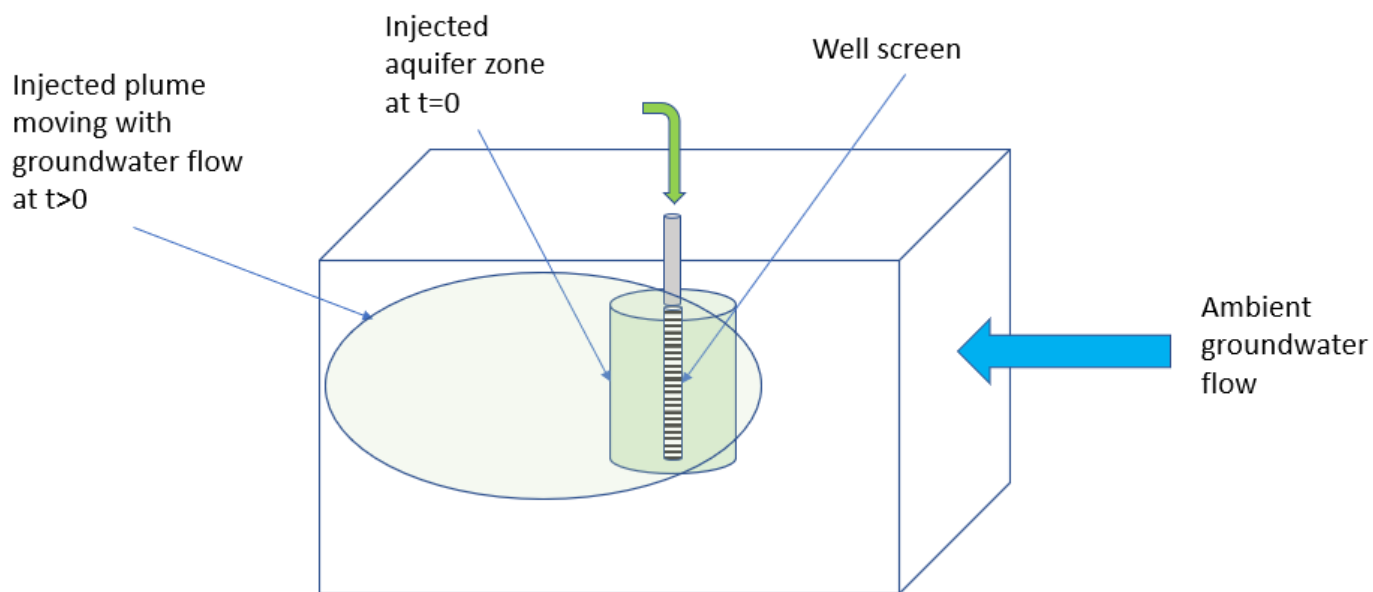


1 inch = 3,000 feet



Site Plan
Stringfellow Superfund Site
Jurupa Valley, California

Figure
2-2



Injected water is assumed to move as "plug flow" through the aquifer, displacing groundwater in mobile (effective) porosity.

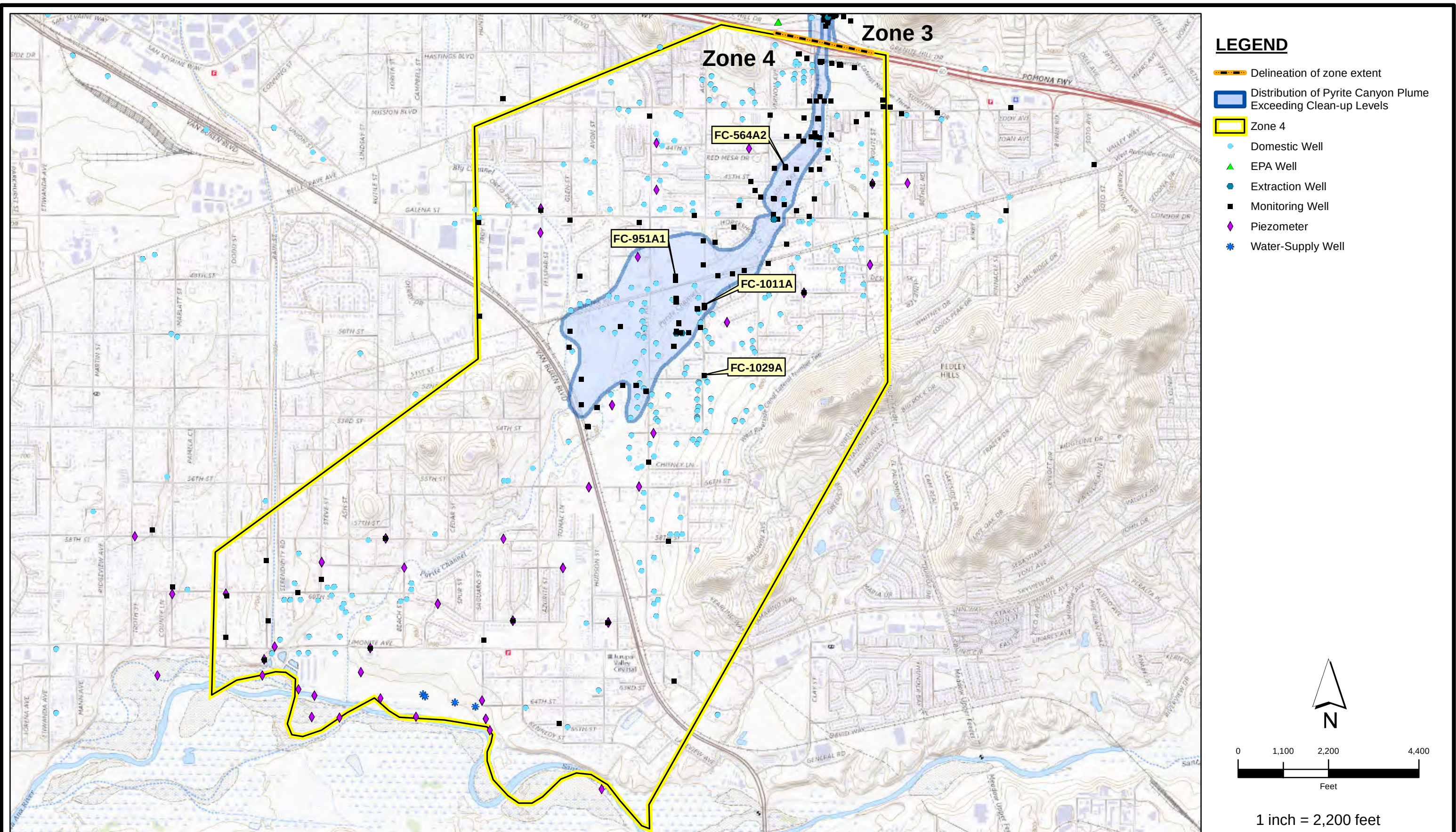


Figure 4-1

Single Well Natural
Gradient Tracer Test
Schematic

PROJECT NO.
500520

DATE:
Apr 2022



Zone 4 Site Wells
Stringfellow Superfund Site
Jurupa Valley, California

Figure
4-2



LEGEND

- Distribution of Pyrite Canyon Plume Exceeding Clean-up Levels
- Domestic Well
- Extraction Well
- Monitoring Well
- Piezometer

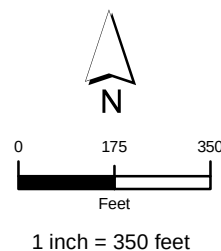
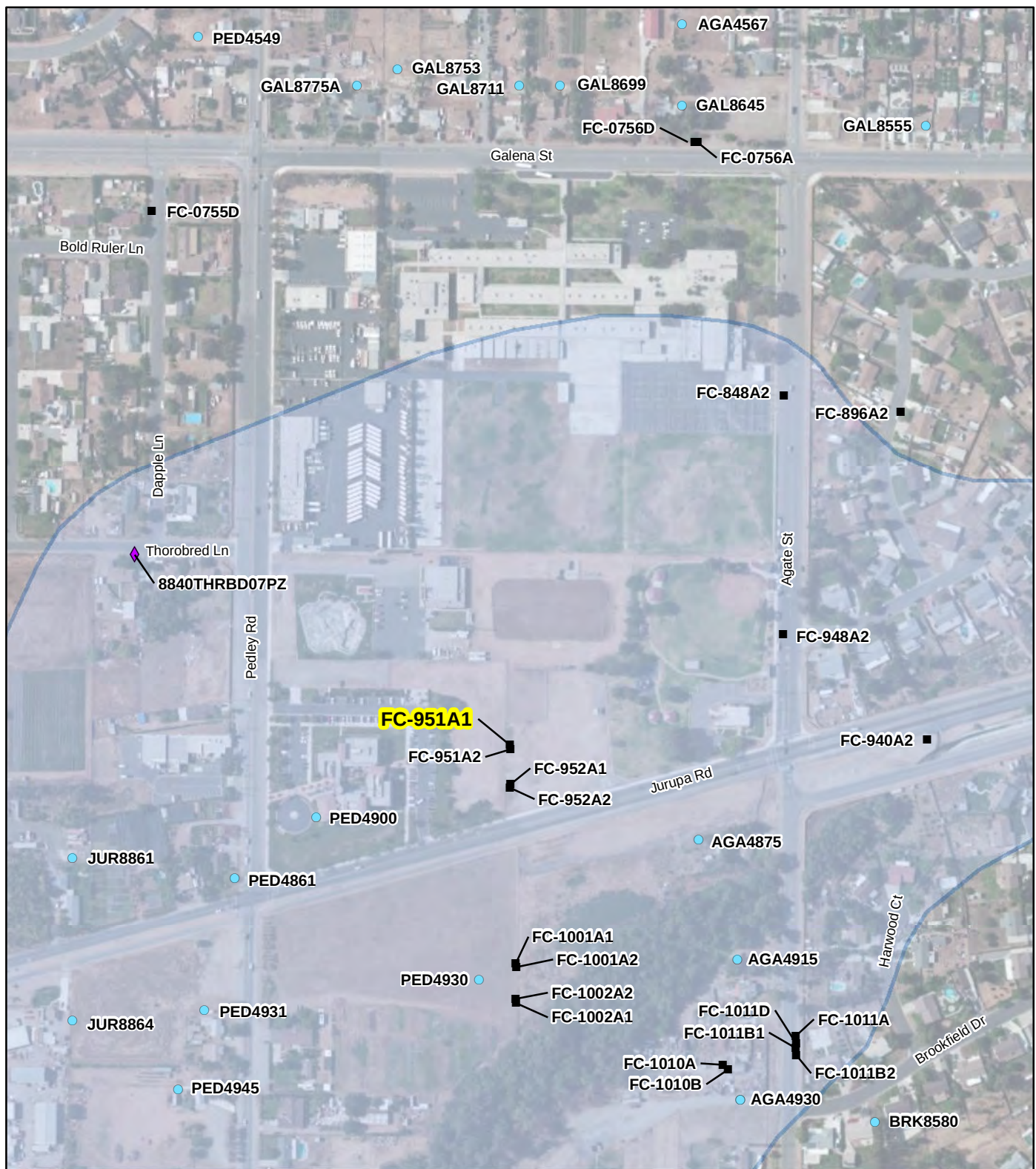


Figure 4-3

Well FC-564A2 Location

Stringfellow Superfund Site
Jurupa Valley, California



LEGEND

- Distribution of Pyrite Canyon Plume Exceeding Clean-up Levels
- Domestic Well
- Monitoring Well
- ◆ Piezometer

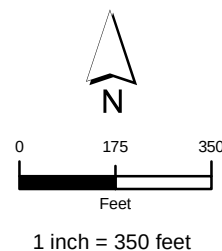
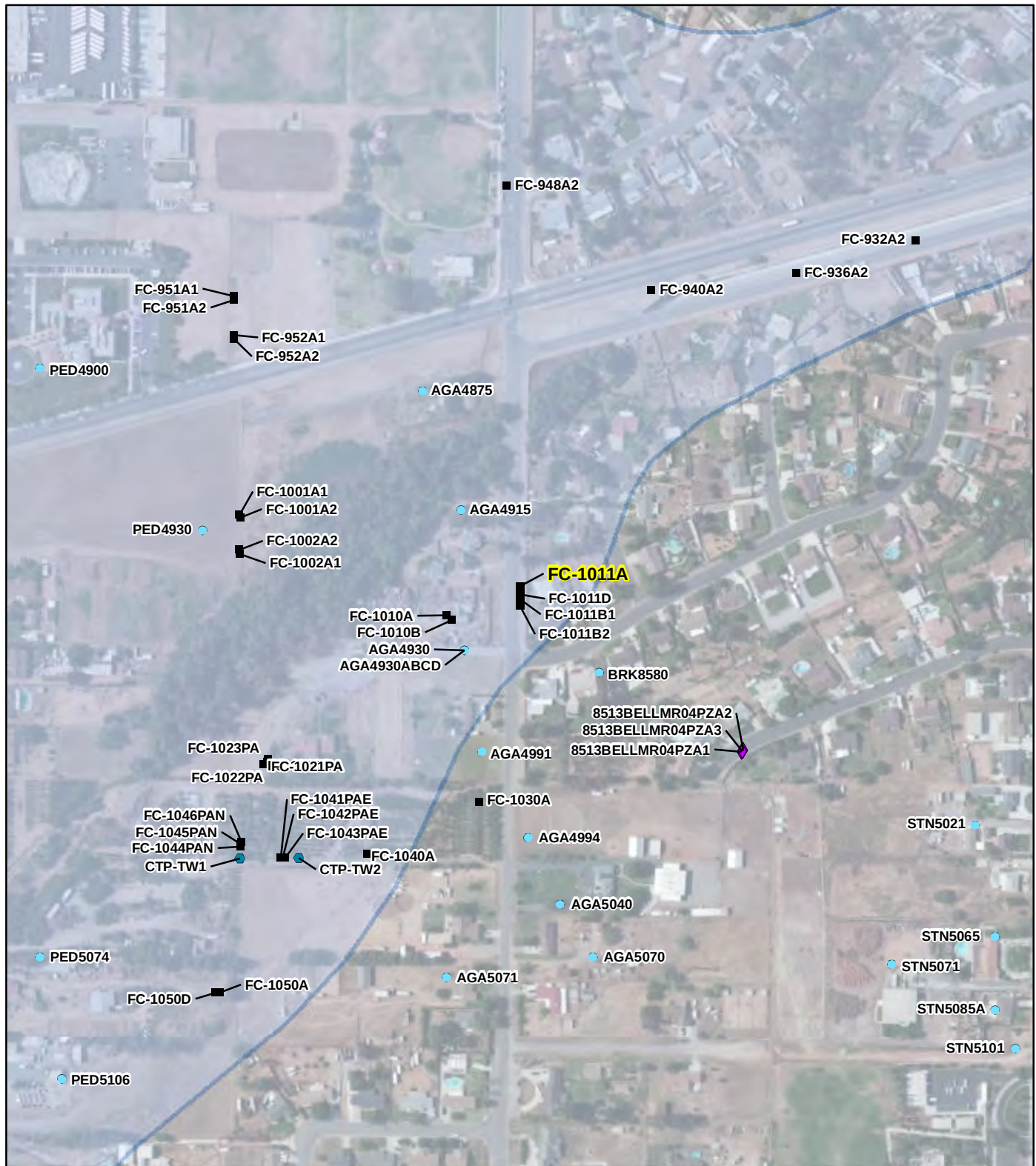


Figure 4-4

Well FC-951A1 Location

Stringfellow Superfund Site
Jurupa Valley, California



LEGEND

- Distribution of Pyrite Canyon Plume Exceeding Clean-up Levels
- Domestic Well
- Extraction Well
- Monitoring Well
- ◆ Piezometer

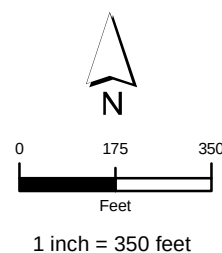


Figure 4-5

Well FC-1011A Location

Stringfellow Superfund Site
Jurupa Valley, California

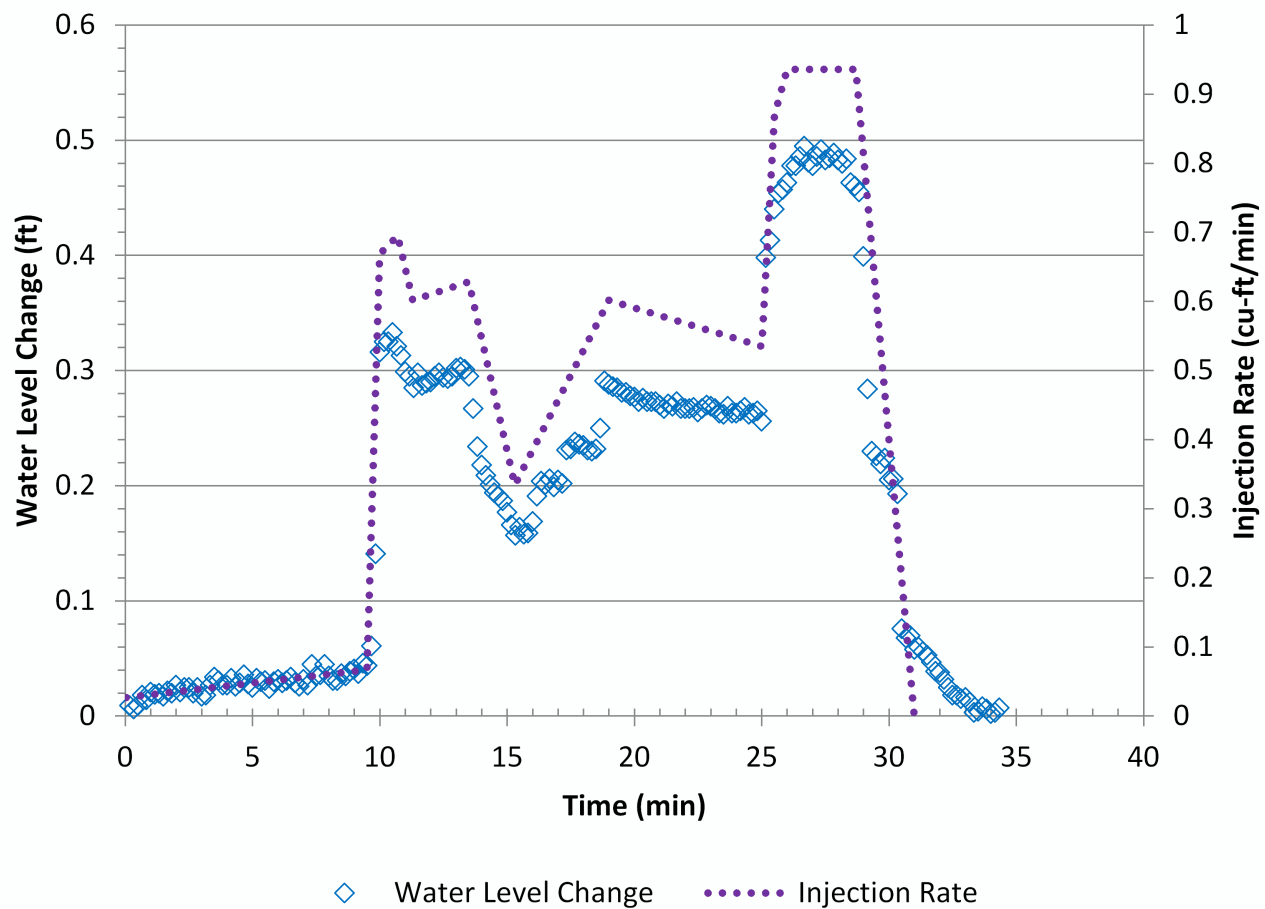


Figure 4-7

Well FC564A2 Injection

PROJECT NO.
500520

DATE:
Apr 2022

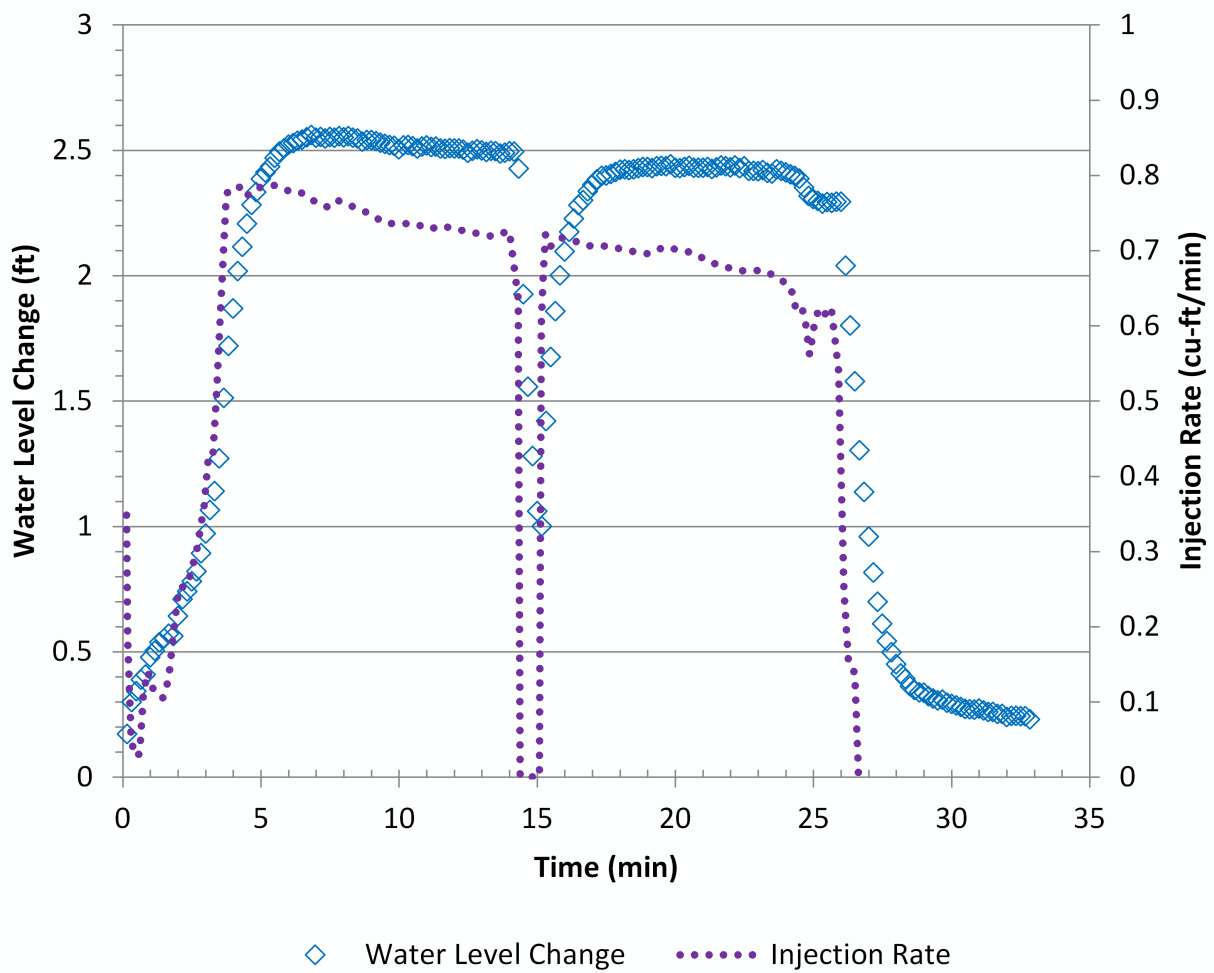


Figure 4-8

Well FC951A1 Injection

PROJECT NO.
500520

DATE:
Apr 2022

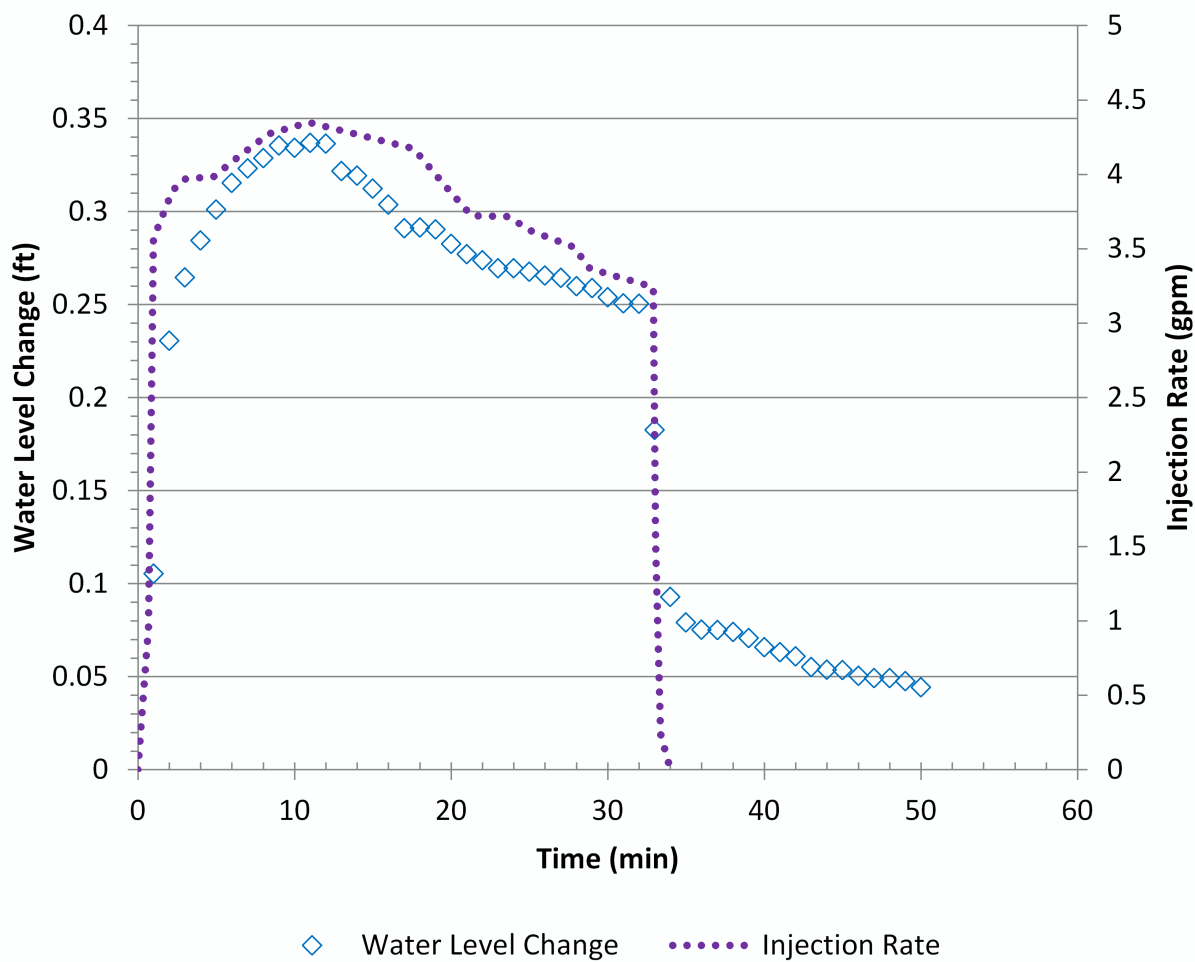


Figure 4-9

Well FC1011A Injection

PROJECT NO.
500520

DATE:
Dec 2022

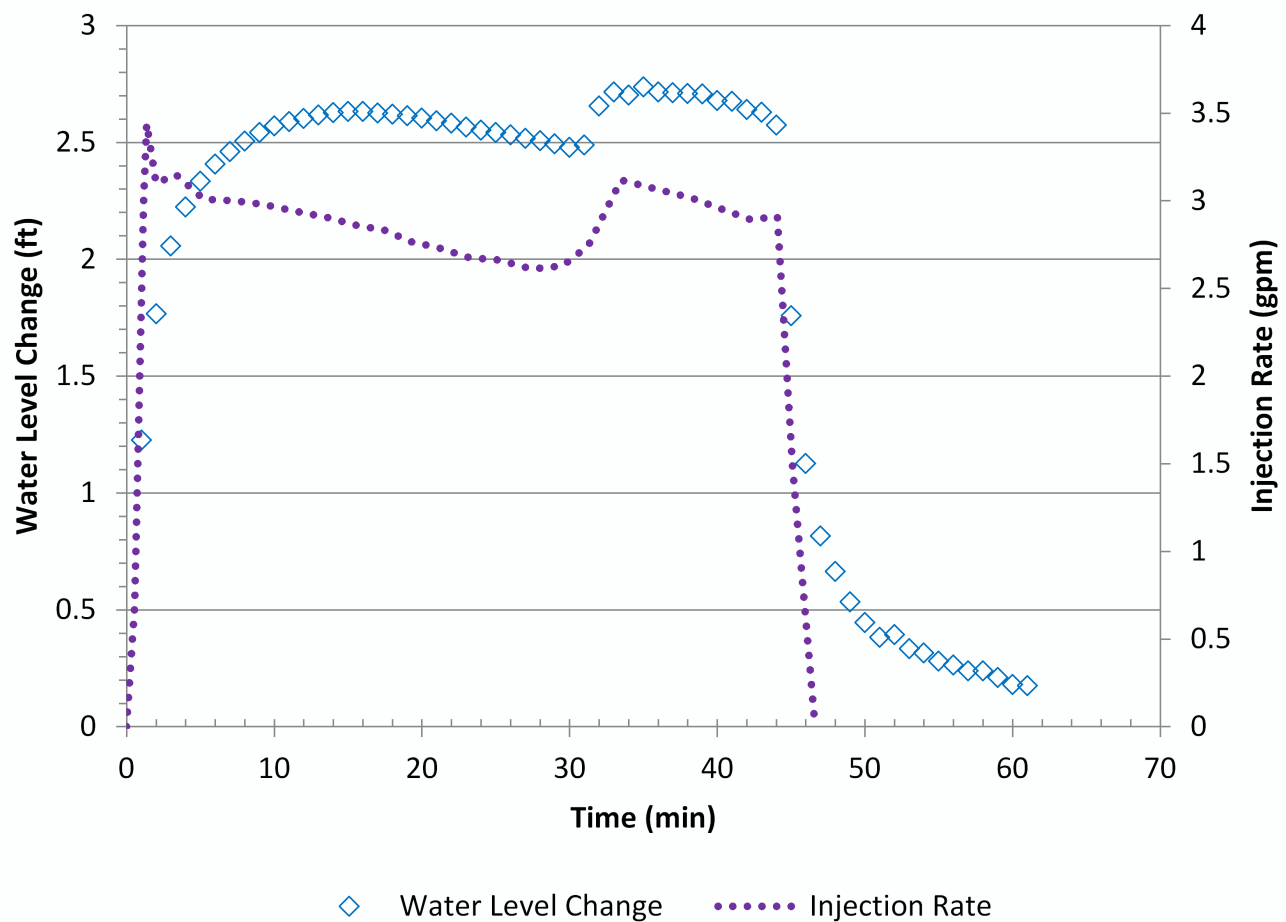


Figure 4-10

Well FC1029A Injection

PROJECT NO.
500520

DATE:
Dec 2022

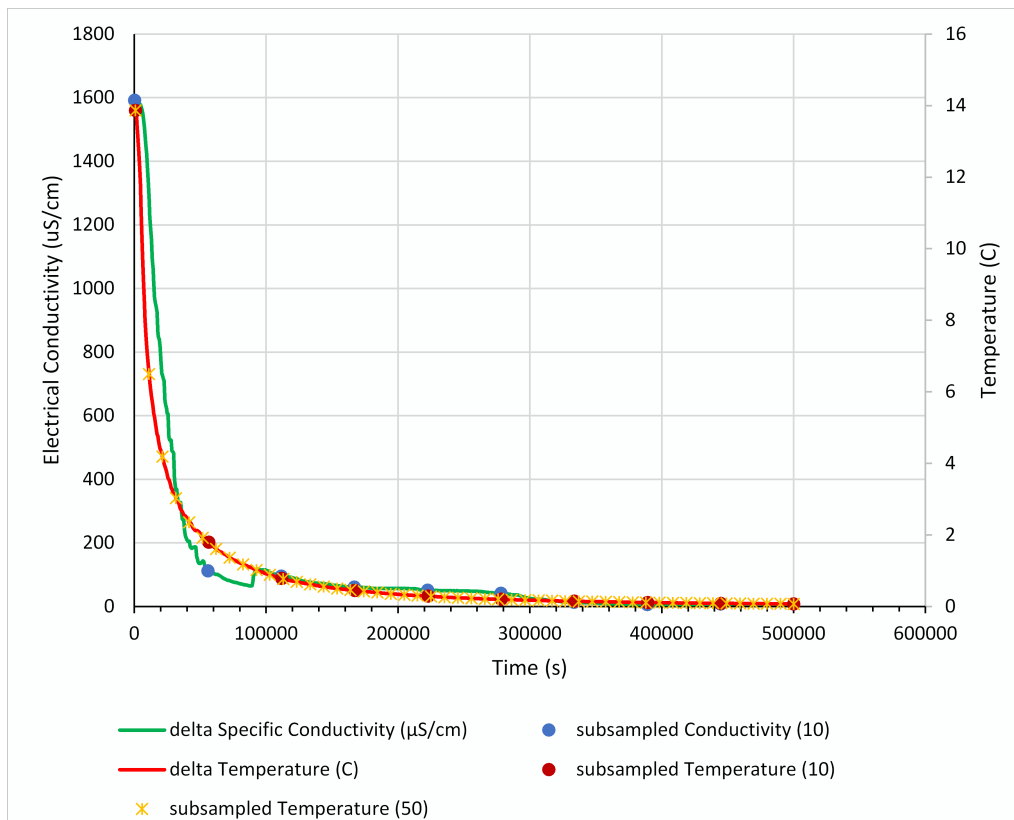
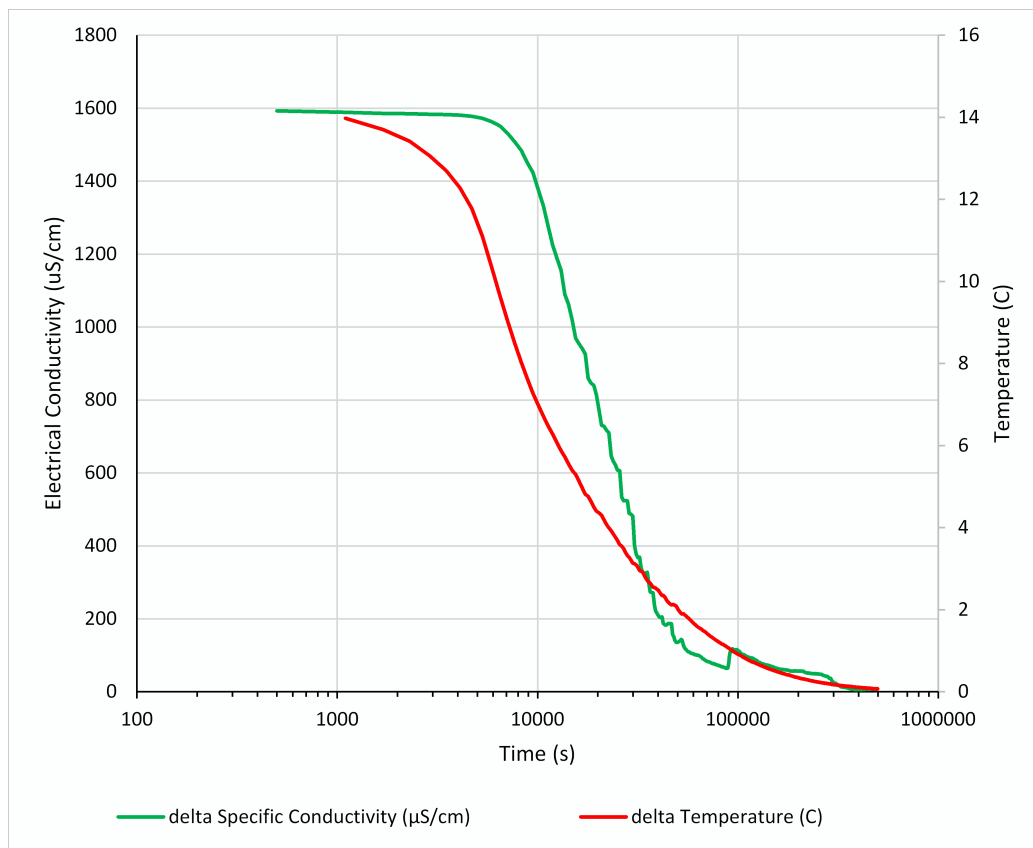


Figure 4-11

Well FC564A2 Tracer Data

PROJECT NO.
500520

DATE:
Apr 2022

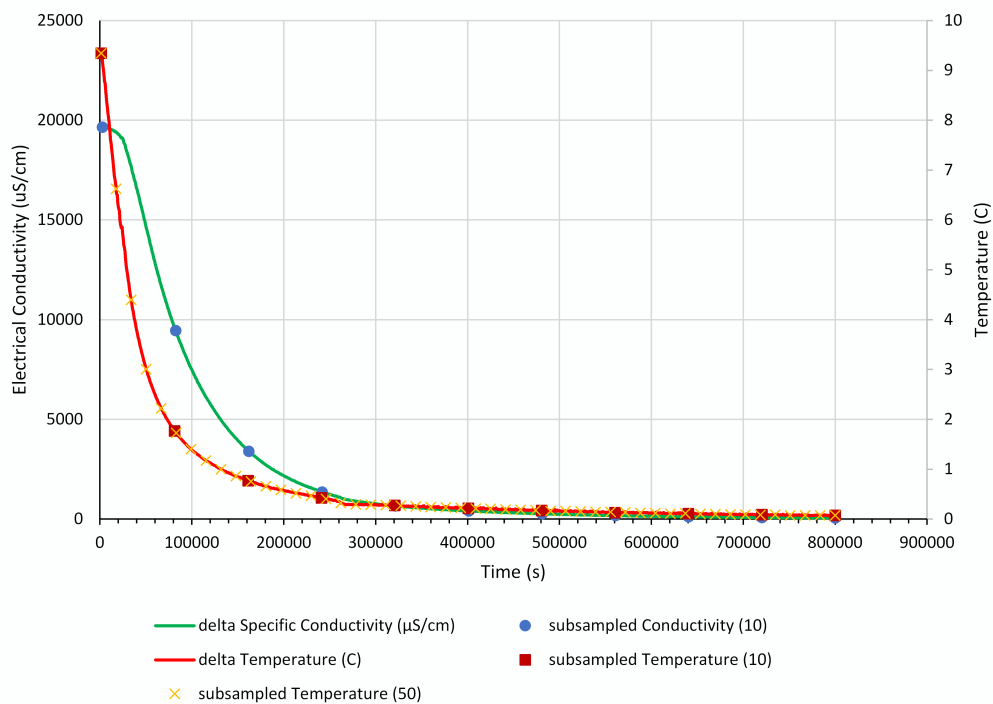
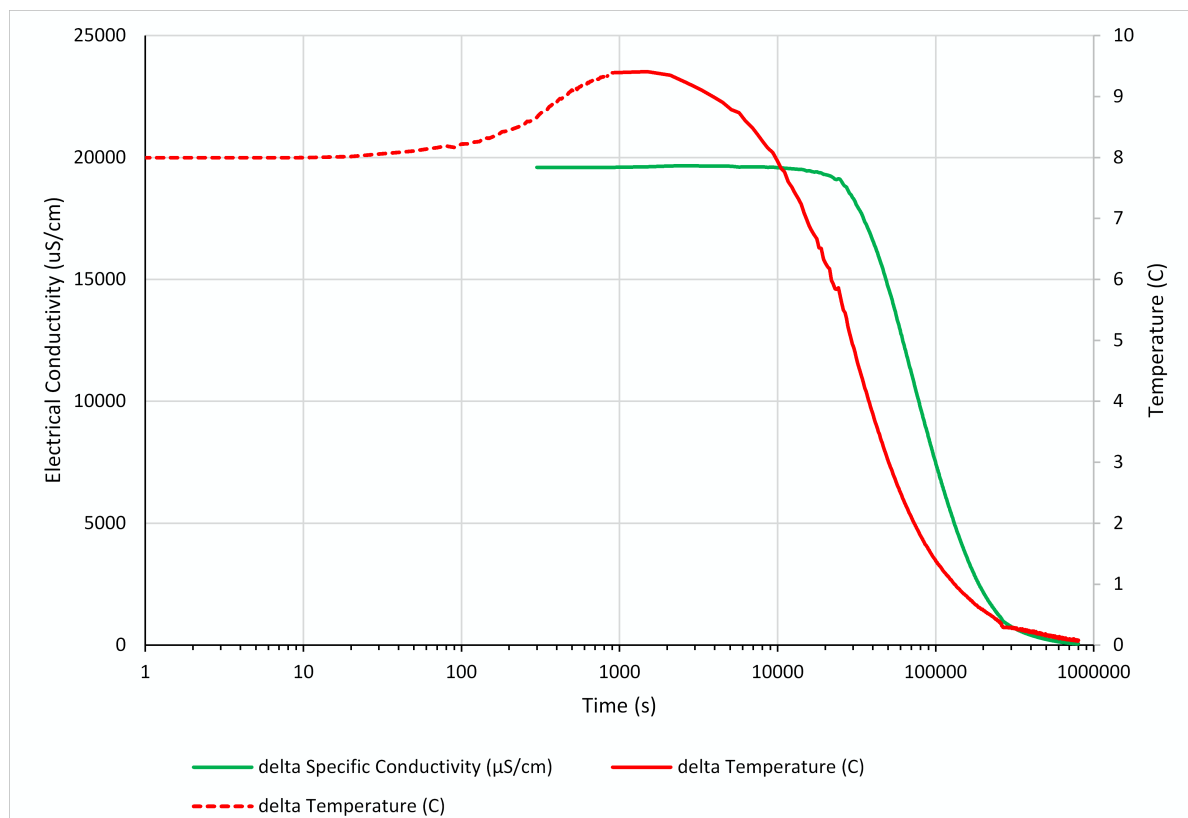


Figure 4-12

Well FC951A1 Tracer Data

PROJECT NO.
500520

DATE:
Apr 2022

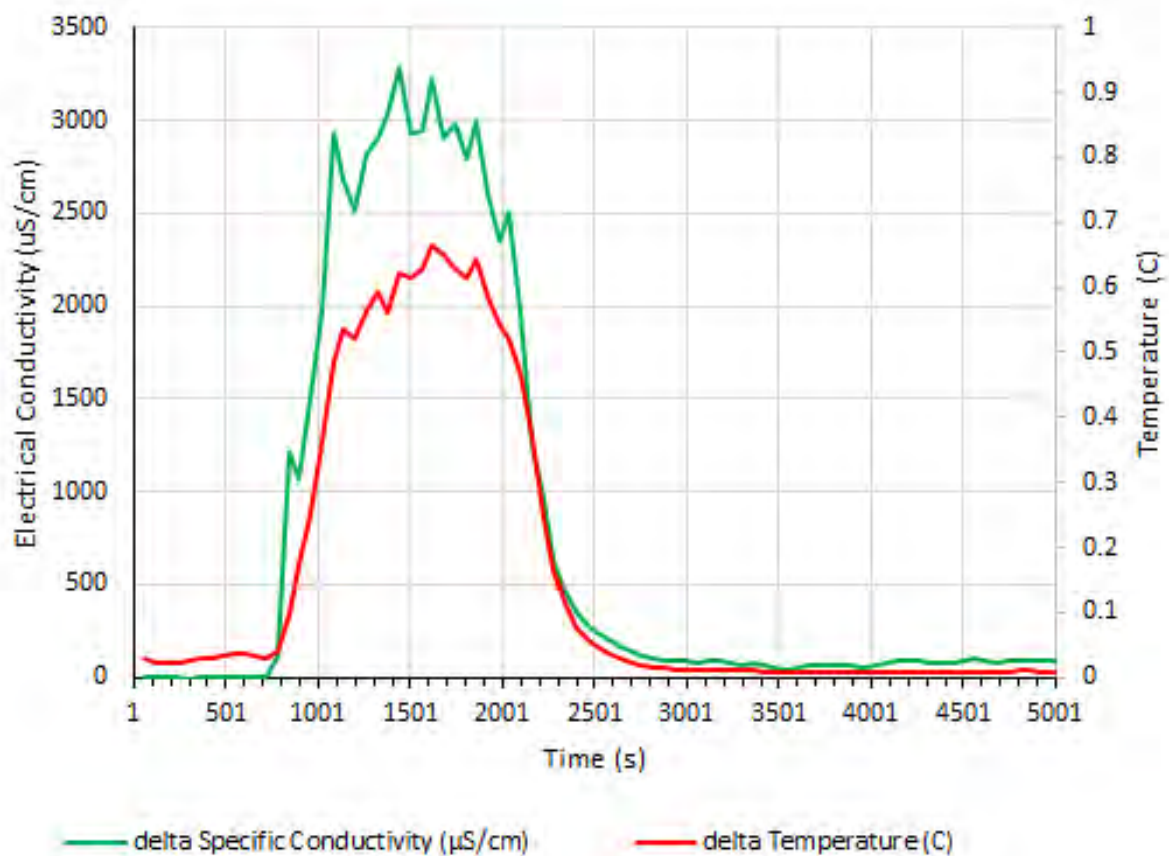


Figure 4-13

Well FC1011A Tracer Data

PROJECT NO.
500520

DATE:
Dec 2022

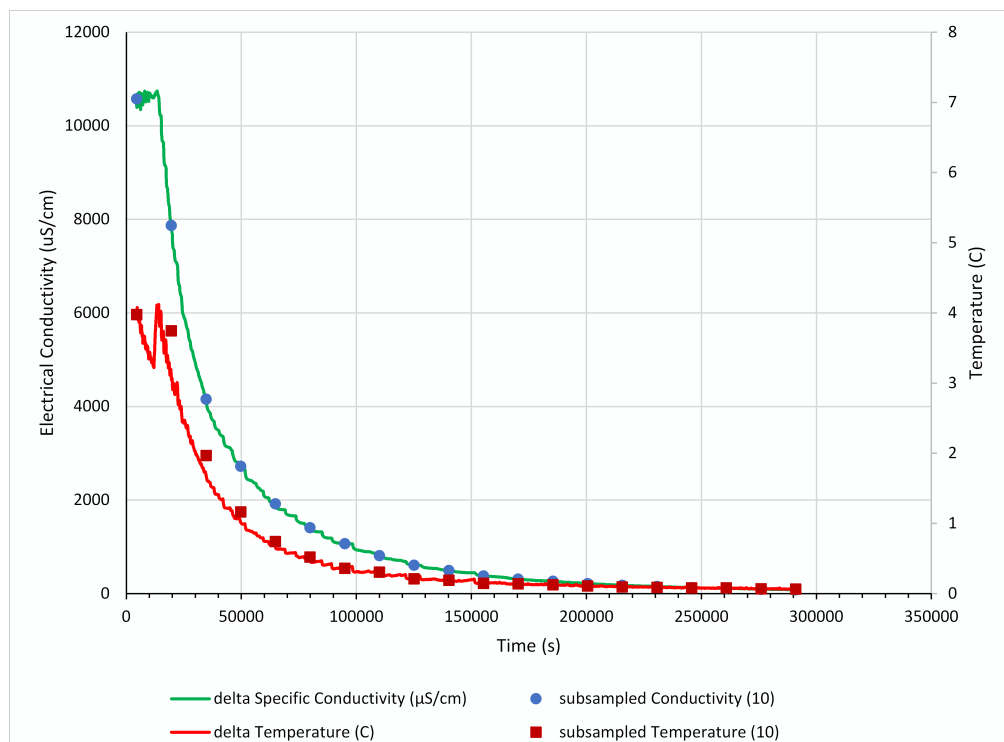
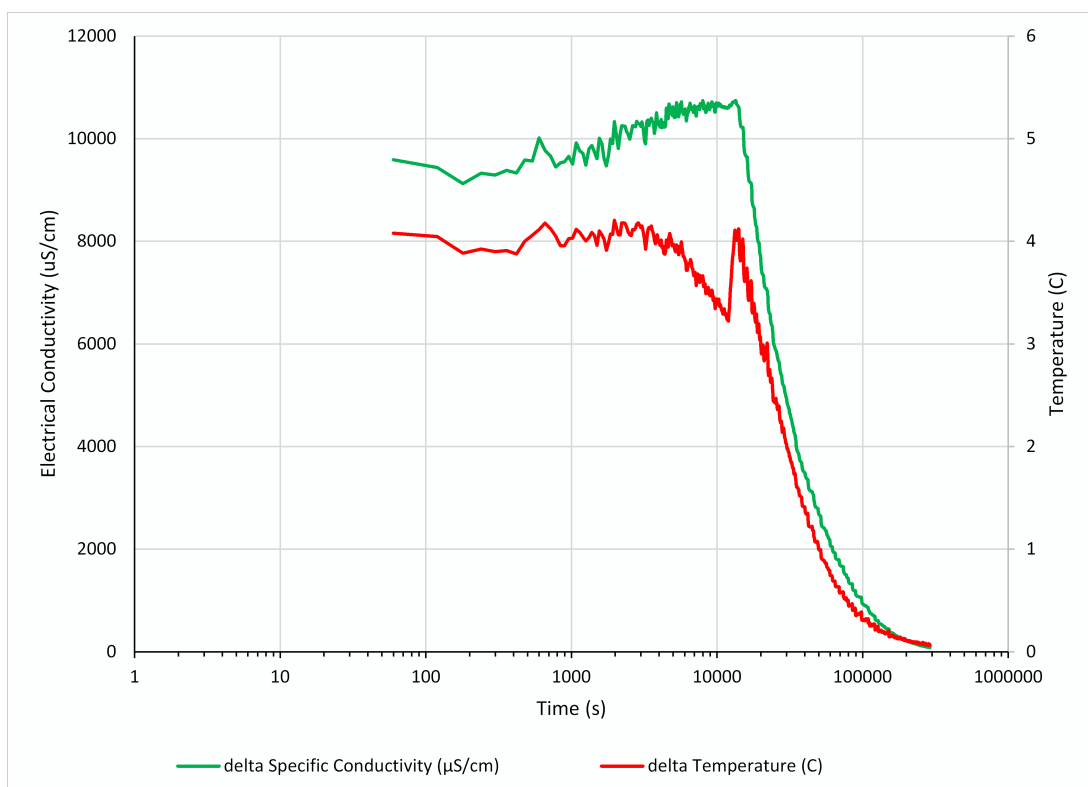


Figure 4-14

Well FC1029A Tracer Data

PROJECT NO.
500520

DATE:
Dec 2022

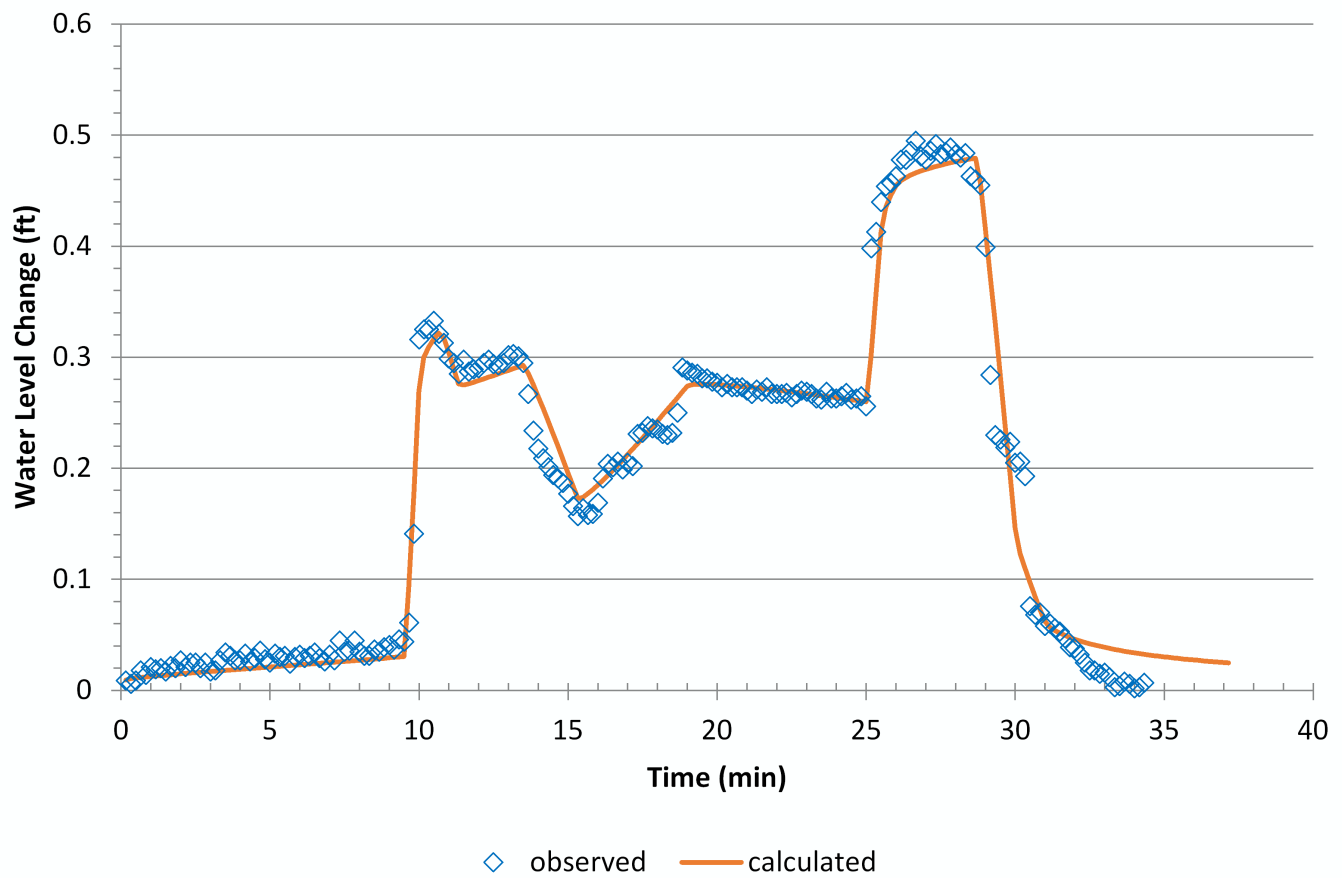


Figure 5-1

FC564A2 Injection
Hydraulic Analysis

PROJECT NO.
500520

DATE:
Apr 2022

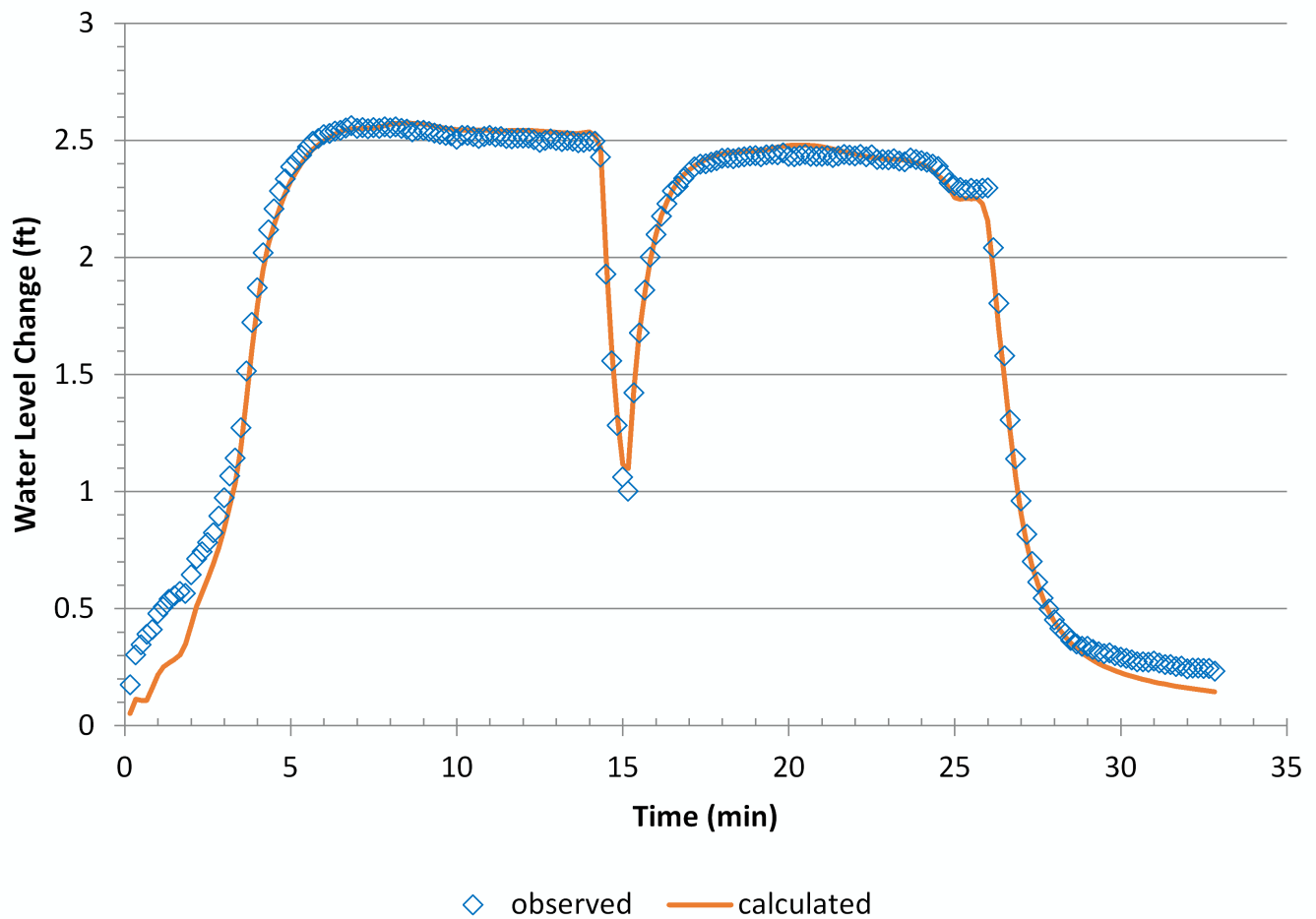


Figure 5-2

FC951A1 Injection
Hydraulic Analysis

PROJECT NO.
500520

DATE:
Apr 2022

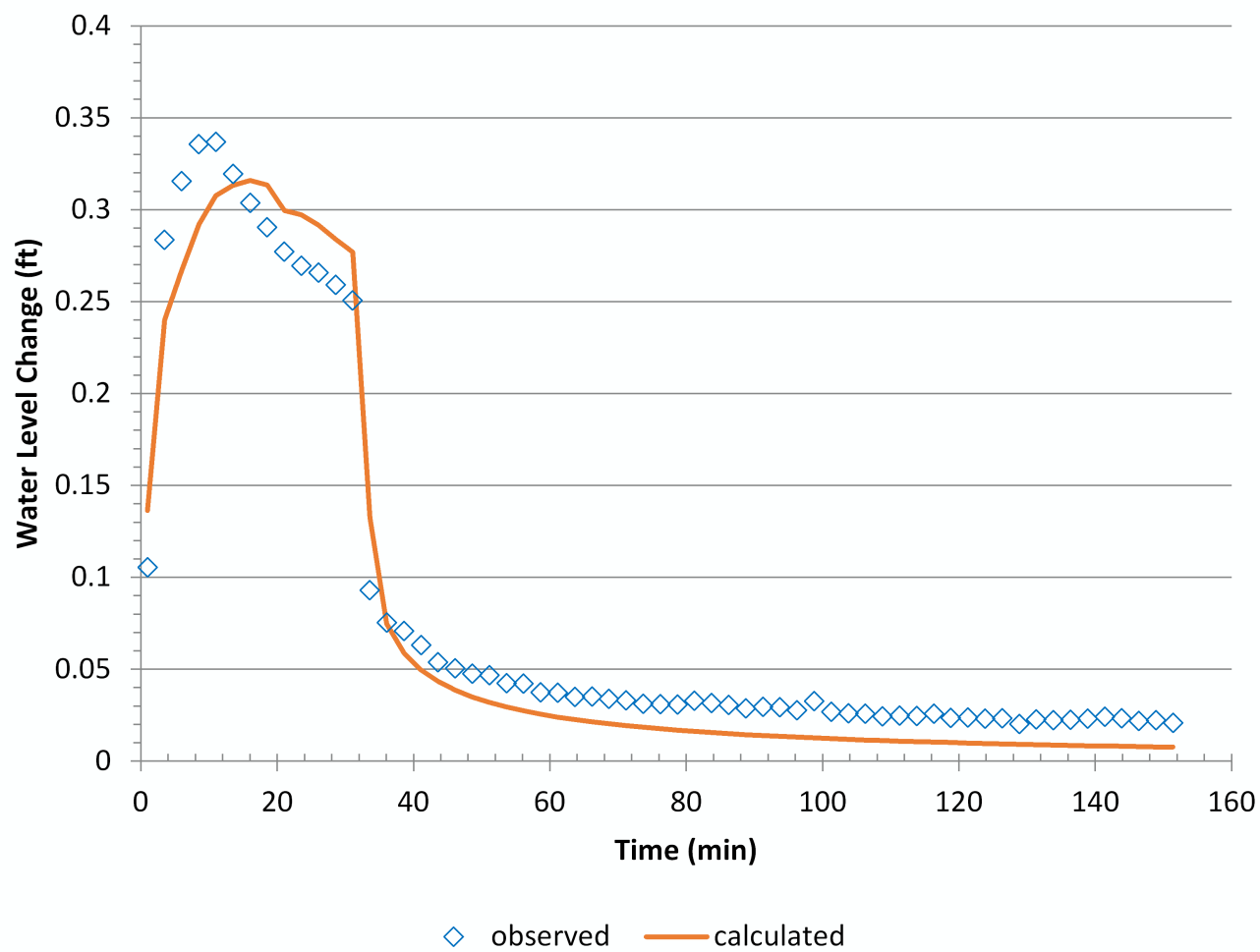


Figure 5-3

FC1011A
Injection Hydraulic
Analysis

PROJECT NO.
500520

DATE:
Dec 2022

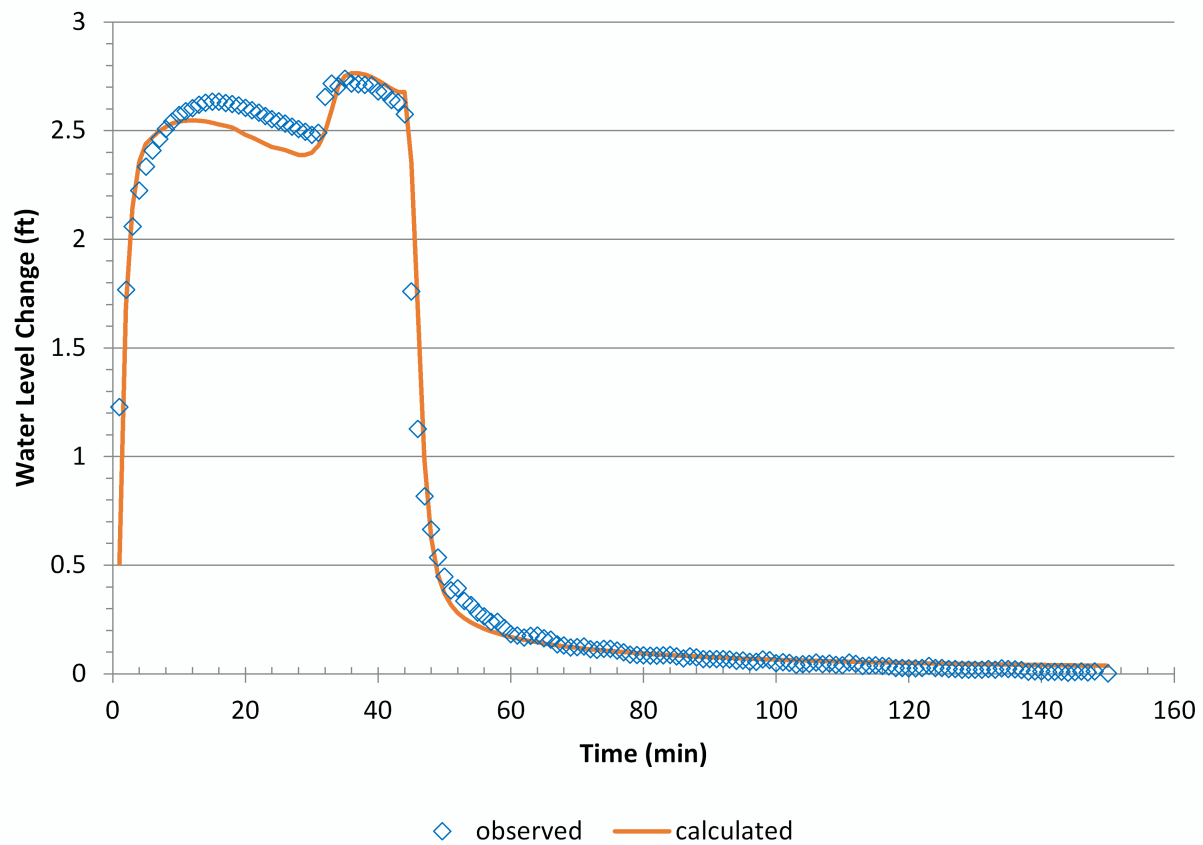


Figure 5-4

FC1029A
Injection Hydraulic
Analysis

PROJECT NO.
500520

DATE:
Dec 2022

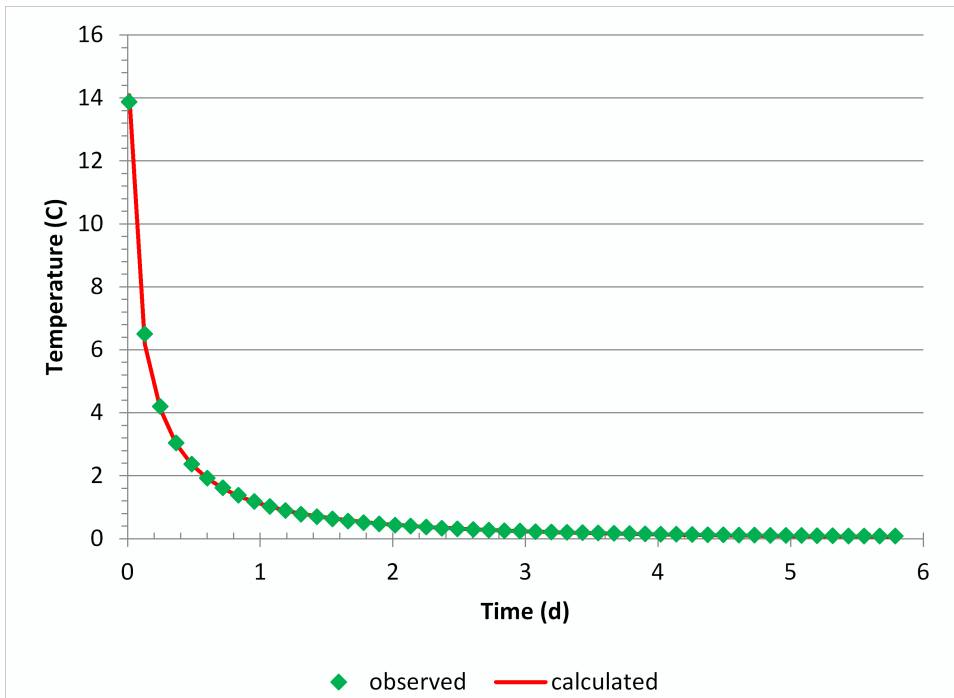
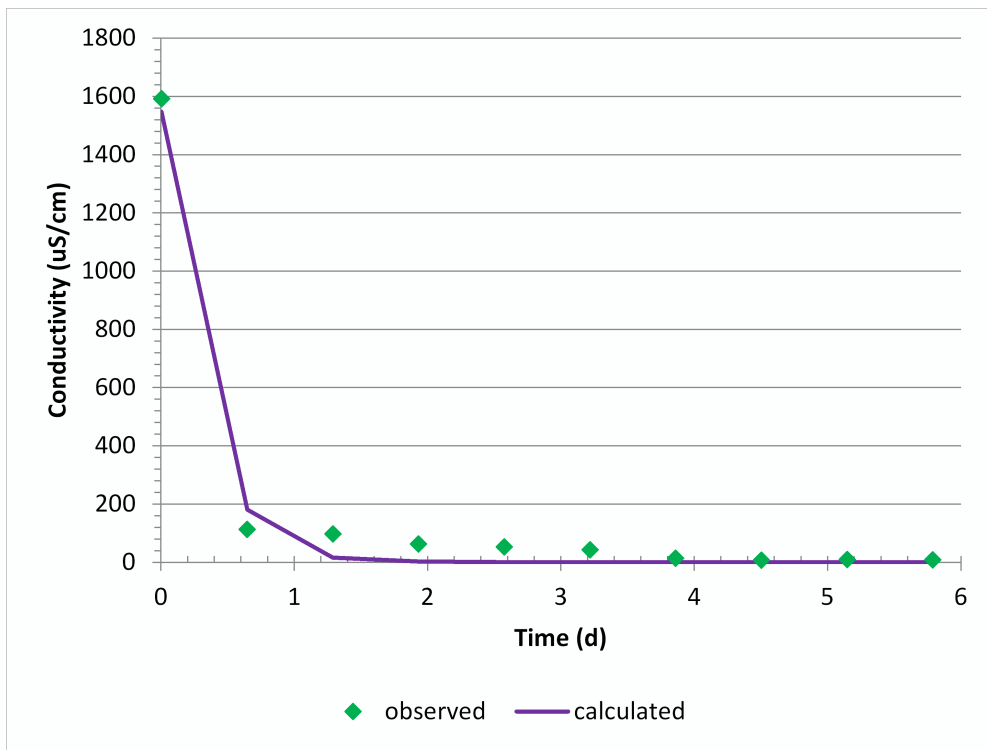


Figure 5-5

Model Fit to FC564A2
Tracer Test
Observations

PROJECT NO.
500520

DATE:
Apr 2022

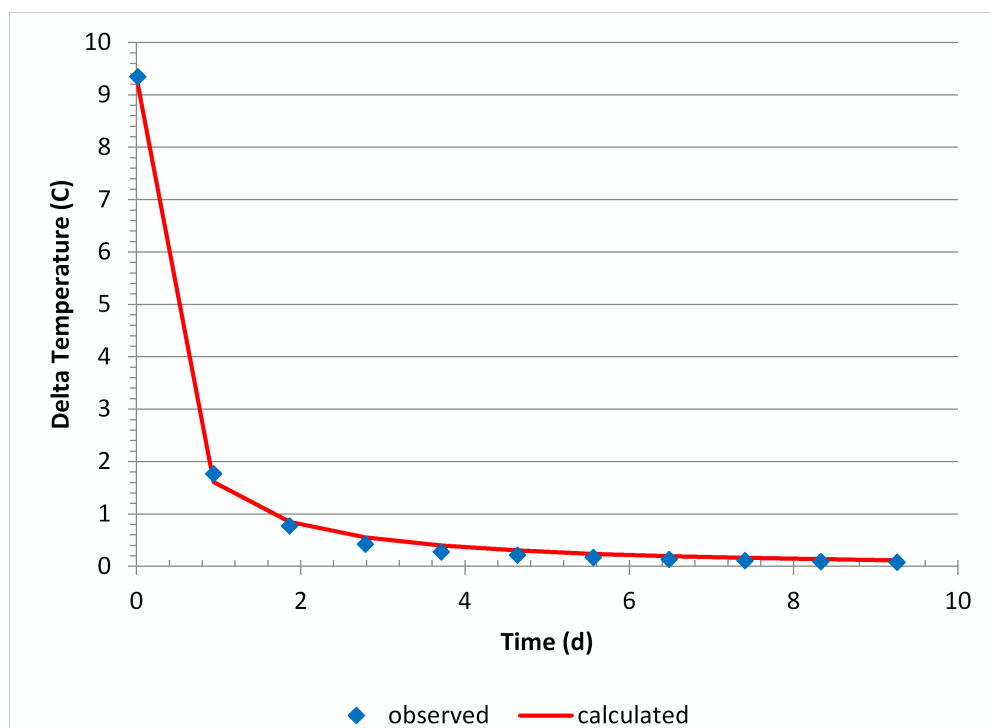
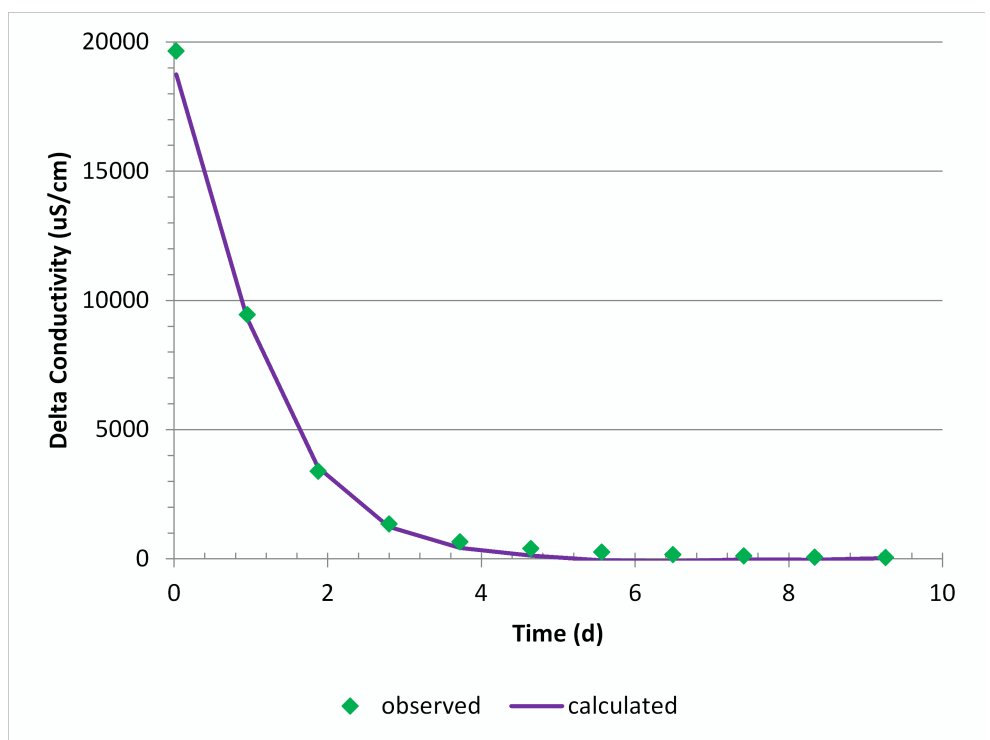


Figure 5-6

Model Fit to FC951A1
Tracer Test
Observations

PROJECT NO.
500520

DATE:
Apr 2022

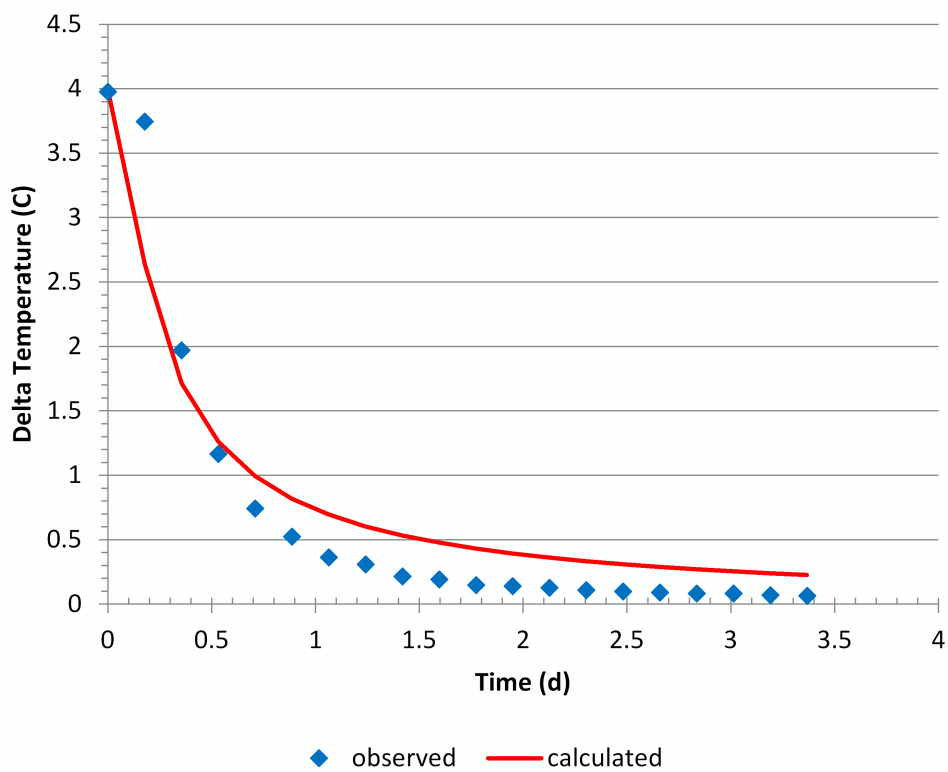
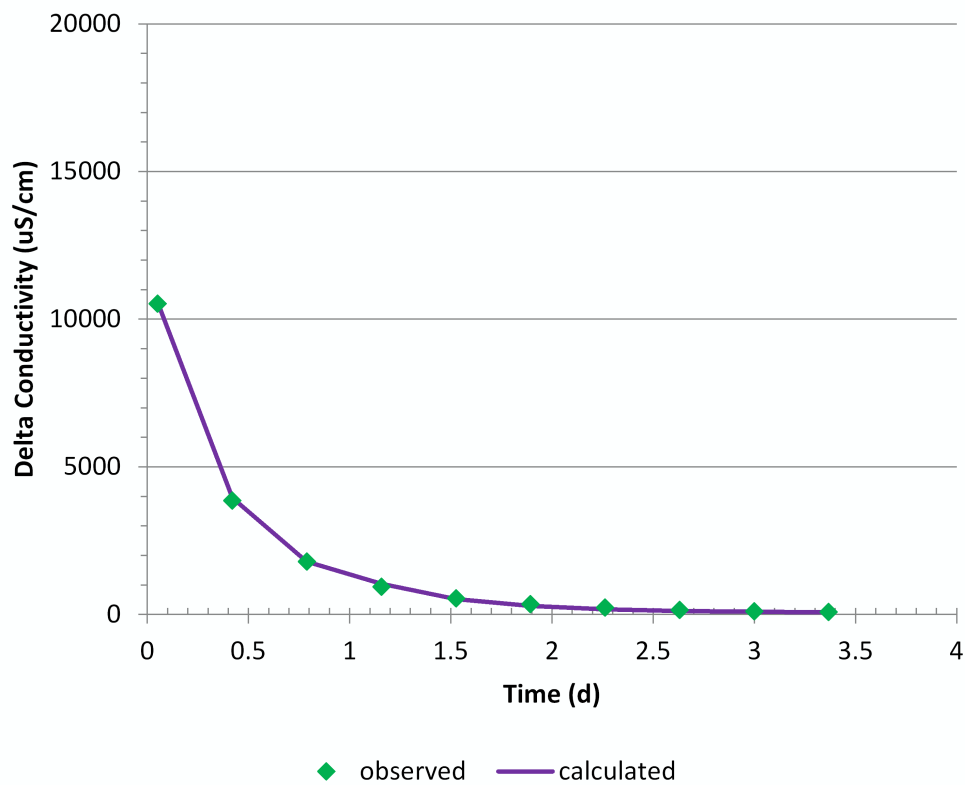


Figure 5-7

Model Fit to
FC1029A Tracer Test
Observations

PROJECT NO.
500520

DATE:
Dec 2022

Tables

Table 5-1
Well Construction

	FC951A1	FC564A2	FC1011A	FC1029A
ztop	3.63	0	0	0
zbot	8.63	16.64	42.66	28
rc	0.24	0.226	0.156	0.238
rw	0.583	0.324	0.156	0.238
rs	0.9	0.6	0.5	0.5
b	27.3	21.12 (96.12)	47.66	35

Notes:

All values are in feet.

Values for rc,rw, b are from Environ 2015.

rc - casing radius

rw - screen radius

rs - skin radius (assumed), not used

b - aquifer thickness

ztop - depth to top of screen below top of aquifer

zbot - depth to bottom of screen below top of aquifer

Table 5-2
Hydraulic Analysis Results

Parameter	Units	FC951A1	FC564A2(a)	FC564A2(b)	FC564A2(c)	FC1011A	FC1011A	FC1029A	FC1029A
Kr	ft/min	3.45E-02	1.42E-01	1.47E-01	1.20E-01	3.10E-02	3.14E-02	8.68E-03	6.47E-03
Ss	ft ⁻¹	1.12E-04	3.43E-06	2.83E-06	1.17E-05	6.17E-04	3.42E-03	4.63E-06	2.28E-07
Kz/Kr	-	1.57E-02	1.03E-01	2.77E-02	2.61E-02	1.99E-04	8.04E-05	1.17E-03	1.31E-01
Krs	ft/min					1.40E-01		1.61E-03	
R ²	-	0.996	0.978	0.977	0.978	0.979	0.979	0.991	0.991
Kr	ft/d	50	204	211	172	45	45	12	9
Kr - slug test (Environ 2015)	ft/d	45	71	71	71	40	40	3.3	3.3
Kr - 1985 pumping test (Environ 2015)	ft/d	57	80	80	80	-	-	-	-

Notes:

a - Manually revised injection rates

b - Manually revised injection rates and effective casing diameter

c - Changes as in b and saturated thickness includes A + WB (96ft)

Kr results for slug tests and pumping tests are from Environ 2015

R - Pearson correlation coefficient

A - Alluvium

WB - Weathered bedrock

Kr - Hydraulic conductivity in radial direction

Kz - Hydraulic conductivity in vertical direction

Kz/Kr - Hydraulic conductivity ratio

Krs - Hydraulic conductivity of well skin in radial direction

Ss - Specific storage

Table 5-3
Transport Analysis Results

Model	Simulation	FC951A1																			
		C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	C0a	C0b	C3a	C3b	C4a	C4b	C5a	C5b	C7a	C7b
		-	4	4	2	4	4	2	-	2	2	-	-	2	2	4	4	4	4	-	-
	Heat	4	4	4	-	4	-	-	4	4	4	4	4	-	-	4	4	-	-	4	4
<i>Estimated parameters</i>																					
q	(m/s)	3.57E-06	4.22E-07	6.69E-07	3.75E-07	1.01E-06	7.58E-07	2.41E-07	2.20E-06	1.15E-06	5.54E-07	4.71E-06	4.31E-06	3.50E-07	1.02E-06	8.69E-07	1.14E-06	6.52E-07	5.25E-07	2.14E-06	2.20E-06
alpha	(s ⁻¹)	-	-	-	2.08E-07	-	-	6.26E-08	-	4.27E-06	9.26E-08	-	-	2.06E-07	3.53E-06	-	-	-	-	-	-
phi	(-)	1.73E-01	5.24E-01	4.75E-01	1.45E-01	5.09E-01	2.99E-01	1.81E-01	3.85E-01	7.48E-01	6.39E-01	1.11E-01	1.21E-01	1.53E-01	6.05E-01	4.35E-01	5.50E-01	2.59E-01	1.22E-01	3.56E-01	4.37E-01
ax	(m)	1.44E-02	1.04E+00	2.00E-01	1.76E-01	1.49E-01	1.67E-01	3.98E-01	3.78E-02	3.70E-02	2.22E-01	1.98E-01	1.39E-01	1.91E-01	8.48E-03	1.76E-01	1.59E-01	1.96E-01	2.82E-01	4.05E-02	4.36E-02
theta	(-)	-	5.14E-02	0.17	-	-	-	-	0.10	-	0.10	-	-	-	-	-	-	-	-	-	-
N	Mass	-	11	11	11	11	11	11	-	11	11	-	-	-	-	-	-	-	-	-	-
N	Heat	50	50	11	-	11	-	-	11	11	11	50	50	11	11	11	11	11	11	-	-
<i>Parameter held constant</i>																					
theta	(-)	0.35	-	-	0.35	0.35	0.35	-	0.35	0.35	-	0.3	0.4	0.3	0.4	0.3	0.4	0.3	0.4	0.3	0.4
phi	(-)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
<i>Calculated value</i>																					
v	(m/s)	7.09E-04	1.56E-05	8.34E-06	7.37E-06	5.69E-06	7.25E-06	1.33E-05	1.63E-05	4.41E-06	8.56E-06	7.95E-05	7.73E-05	7.63E-06	4.22E-06	1.64E-05	1.80E-05	1.11E-05	4.66E-06	1.76E-04	1.26E-04
v	(ft/d)	201.1	4.4	2.4	2.1	1.6	2.1	3.8	4.6	1.2	2.4	22.5	21.9	2.2	1.2	4.7	5.1	3.1	1.3	49.9	35.7
q	(ft/d)	1.013	0.120	0.190	0.106	0.288	0.215	0.068	0.624	0.327	0.157	1.335	1.221	0.099	0.289	0.246	0.325	0.185	0.149	0.607	0.623
thetam	(-)	0.060	0.027	0.080	0.051	0.178	0.104	0.018	0.135	0.262	0.065	0.033	0.048	0.046	0.242	0.131	0.220	0.078	0.049	0.107	0.175
ax	(ft)	4.72E-02	3.43E+00	6.56E-01	5.77E-01	4.90E-01	5.46E-01	1.30E+00	1.24E-01	1.21E-01	7.27E-01	6.48E-01	4.57E-01	6.26E-01	2.78E-02	5.78E-01	5.21E-01	6.42E-01	9.25E-01	1.33E-01	1.43E-01
alpha	(d ⁻¹)	-	-	-	1.80E-02	-	-	5.41E-03	-	3.69E-01	8.00E-03	-	-	1.78E-02	3.05E-01	-	-	-	-	-	-

Notes:
q - Darcy flux
alpha - Mass transfer rate between mobile and immobile porosity
phi - Fraction of total porosity that is mobile domain
ax - Longitudinal dispersivity
theta - Total porosity (equal to water content for saturated soil)
v - Advective velocity
thetam - Mobile porosity
N - Number of observations (sub-sampled)
Transverse dispersivities were specified as tied parameters, defined as 0.1 and 0.01 of longitudinal dispersivity.
Light shaded results are considered plausible.
Dark shaded results are considered representative.
Unshaded results are considered implausible because theta is too low or q is too high.

Table 5-3
Transport Analysis Results

		FC564A2												FC1029A					
Simulation		A1	A2	A3	A4	A5	A6	A7	A8	A7a	A7b	A4a	A4b	B0	B1	Cd0	C2	T0	T1
Mass		4	4	4	-	2	2	2	2	2	2	-	-	2	2	2	4.1	-	-
Heat	Model	4	4	-	4	-	-	4	4	4	4	4	4	4	4	-	-	4.1	4
<i>Estimated parameters</i>																			
q	(m/s)	2.98E-06	1.57E-06	1.47E-06	2.66E-06	1.76E-06	3.06E-06	2.86E-06	3.12E-06	2.61E-06	3.14E-06	2.51E-06	2.96E-06	8.53E-07	6.10E-07	3.40E-07	2.80E-07	8.45E-06	9.82E-06
alpha	(s ⁻¹)					5.05E-06	1.56E-05	8.08E-06	1.77E-05	5.14E-06	1.29E-05			2.422E-07	3.899E-07	2.062E-07			
phi	(-)	4.63E-01	2.75E-01	2.58E-01	6.87E-01	1.38E-01	3.39E-01	7.37E-01	7.00E-01	7.18E-01	7.51E-01	8.19E-01	8.24E-01	1.99E-01	2.36E-01	6.99E-02	1.82E-02	6.91E-02	
ax	(m)	2.76E-01	1.35E+00	1.97E+00	1.02E-01	2.74E-01	6.14E-01	3.06E-02	4.43E-02	9.04E-03	5.43E-02	2.67E-03	6.04E-02	2.77E-01	2.35E-01	7.76E-01	5.66E-01	3.11E-02	1.35E-02
theta	(-)		0.33				0.44		0.40						0.19				2.26E-02
N	Mass	10	10	10		10	10	10	10					10	10	10	20		
N	Heat	10	10		10			50	50			50	50	20	20			20	20
<i>Parameter held constant</i>																			
theta	(-)	0.35		0.35	0.35	0.35		0.35		0.3	0.4	0.3	0.4	0.35		0.35	0.35	0.35	0.5
<i>Calculated value</i>																			
v	(m/s)	1.84E-05	1.74E-05	1.63E-05	1.11E-05	3.65E-05	2.04E-05	1.11E-05	1.12E-05	1.21E-05	1.04E-05	1.02E-05	8.97E-06	1.23E-05	1.32E-05	1.39E-05	4.39E-05	3.49E-04	8.71E-04
v	(ft/d)	5.2	4.9	4.6	3.1	10.3	5.8	3.1	3.2	3.4	3.0	2.9	2.5	3.5	3.8	3.9	12.4	99.0	246.8
q	(ft/d)	0.844	0.446	0.416	0.755	0.499	0.868	0.811	0.885	0.741	0.889	0.712	0.838	0.242	0.173	0.096	0.079	2.395	2.784
thetam	(-)	0.162	0.091	0.090	0.240	0.048	0.150	0.258	0.278	0.215	0.300	0.246	0.330	0.070	0.046	0.024	0.006	0.024	0.011
ax	(ft)	9.06E-01	4.43E+00	6.47E+00	3.34E-01	8.99E-01	2.02E+00	1.00E-01	1.45E-01	2.97E-02	1.78E-01	8.77E-03	1.98E-01	9.07E-01	7.72E-01	2.55E+00	1.86E+00	1.02E-01	4.41E-02
alpha	(d ⁻¹)					4.36E-01	1.35E+00	6.98E-01	1.53E+00	4.44E-01	1.12E+00			2.09E-02	3.37E-02	1.78E-02			

Notes:

q - Darcy flux

alpha - Mass transfer rate between mobile and immobile porosity

phi - Fraction of total porosity that is mobile domain

ax - Longitudinal dispersivity

theta - Total porosity (equal to water content for saturated soil)

v - Advective velocity

thetam - Mobile porosity

N - Number of observations (sub-sampled)

Transverse dispersivities were specified as tied parameters, defined as 0.1 and 0.01 of longitudinal dispersivity.

Light shaded results are considered plausible.

Dark shaded results are considered representative.

Unshaded results are considered implausible because theta is too low or q is too high.

Table 5-4
Comparison of Darcy Flux from Tracer Tests and Aquifer Tests

Well	Aquifer Tests			Tracer Tests q (ft/d)
	K (ft/d)	gradient (-)	q (ft/d)	
FC564A2	80	0.008	0.64	(K from Environ 2015) 0.8
FC564A2	200	0.008	1.6	
FC951A1	57	0.008	0.46	(K from Environ 2015) 0.3
FC951A1	50	0.008	0.4	
FC1011A	40	0.008	0.32	(K from Environ 2015) NA
FC1011A	40	0.012	0.47	
FC1029A	3.3	0.008	0.03	(K from Environ 2015) 0.24
FC1029A	10.5	0.019	0.20	

Notes:

Darcy flux (q) was calculated from K and gradient

K - hydraulic conductivity

Gradient is calculated from water level contours on Figure 5-3 of the 2019 Annual Remedy Report

NA - Not available

Appendix A

Field Readings and Photolog

Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA

DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: June 2021

Photograph
No. 1

Site: Zone 1

Photo Date:

6/29/2021

Description:
Heating distilled
water at PCTF.



Photograph
No. 2

Site: 4

Photo Date:

6/22/2021

Description:
Well FC564A2



Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA

DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: June 2021

Photograph No. 3 Site: Zone 4 Photo Date: 6/29/2021	 Jun 29, 2021 3:29:12 PM 8322 Pristine Place Riverside County California
Description: Injection in FC564A2	

Photograph No. 4 Site: Zone 4 Photo Date: 6/29/2021	2:30:31 PM 37.613 Temperature - C 0.100 Pressure - PSI 0.231 Depth - ft
Description: TROLL 200 readings with probe in the tank prior to injection in FC564A2	15.546 Actual Conductivity - μS/cm 12.527 Specific Conductivity - μS/cm 64327.207 Resistivity - ohm-cm
	0.004 Salinity - PSU 0.008 Total Dissolved Solids - ppt 0.993 Water Density - g/cm3

Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA
DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: June 2021

Photograph
No. 5

Site: 4

Photo Date:
6/29/2021

Description:
Example time-
stamped snapshots
of flowmeter that
were used to record
flowrate at
FC564A2



Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA
DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: June 2021

Photograph
No. 6

Site: 4

Photo Date:
6/30/2021

Description:
Well FC951A1



Photograph
No. 7

Site: Zone 4

Photo Date:
6/30/2021

Description:
Injection in FC951A1





Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA

DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: June 2021

Photograph No. 8 Site: 4 Photo Date: 6/29/2021	2:58:08 PM		
	30.174 Temperature - C	0.652 Pressure - PSI	1.503 Depth - ft
	22924.170 Actual Conductivity - μ S/cm	20862.486 Specific Conductivity - μ S/cm	43.622 Resistivity - ohm-cm
Description: FC951A1 – Initial TROLL 200 reading inside well prior to injection	12.687 Salinity - PSU	13.561 Total Dissolved Solids - ppt	1.005 Water Density - g/cm ³

Photograph No. 9 Site: Zone 4 Photo Date: 6/30/2021	3:16:37 PM		
	25.923 Temperature - C	3.038 Pressure - PSI	
	7.014 Depth - ft		
Description: FC951A1 – Initial TROLL 700 reading inside well prior to injection			

Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA

DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: January 2023

Photograph No. 1 Site: Zone 1 Photo Date: 8/3/2022	 Aug 3, 2022 at 3:52:27 PM 5190 Agate St Jurupa Valley CA 92509 United States	
Description: Set-up during FC1029A test		

Photograph No. 2 Site: 4 Photo Date: 8/3/2022	 Aug 3, 2022 3:38:36 PM 8625 Kim Lane Riverside County California	
Description: Well FC1029A		

Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA

DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: January 2023

Photograph
No. 3

Site: Zone 4

Photo Date:
8/3/2022



Description:
Example time-stamped snapshots of flowmeter that were used to record flowrate at FC1029A

Photograph
No. 4

Site: Zone 4

Photo Date:
4/15/2022



Description:
Well FC1011A

Stringfellow – Field Oversight Photolog

Project: 500520.01400550

Location: Stringfellow Zone 4, Glen Avon, CA
DCN: 1914

Prepared by: APTIM Federal Services LLC

Date: January 2023

Photograph No. 5 Site: 4 Photo Date: 8/4/2022	
Description: Set-up during FC1011A test	
Photograph No. 6 Site: 4 Photo Date: 8/4/2022	
Description: Example time-stamped snapshots of flowmeter that were used to record flowrate at FC1029A	

Transcript of field notes

Notes were supplemented by photos of flowmeter and laptop screen with datalogger readings

FC964A2

6/29/2021

14:15 DTW 43.88 ft

TD 62.15 ft

TROLL 200 at ~50ft

TROLL 700 at ~6.6ft

15:28 start injection

16:08 end injection

FC951A1

6/30/2021

14:50 DTW 21.01 ft

TD 32.32 ft

15:00 started TROLL 700 recording with probe in tank, T=30C

15:20 started TROLL 200 recording, first in well, then in tank, then back to well

15:31 started injection



FIELD ACTIVITY DAILY LOG

Daily Log	Date:	7	26	22
	No.			
	Sheet	1	Of	1

PROJECT NAME: Stringfellow PAT Project		Project Number: S00520	
FIELD ACTIVITY SUBJECT: Slug tests, Pre & Post Downhole Vertical Profiling 251A-2		Task number:	
DESCRIPTION OF DAILY ACTIVITIES AND EVENTS,			
0800: Arrive at security gate. Meet Piotr Kostechi (DTSC). Take site safety training. 0830: Meet with Greg Smith & Phil Harte (USGS) 0845: Piotr shows us where to discharge water. Move to 251 well cluster. 0935: Set up at well cluster in church parking lot. Gauge wells 1000: Piotr on site to inform us that no dyes (even food-grade) can be introduced into the wells nor discharged at sump. 1020: CTN TW1 DTW = 64.52' (extraction well) 11.74 gal/min FC 251A2 DTW = 56.41' TOC TD = 90.25' inner diameter 4.5" FC 251A3 DTW = 55.71' TD = inner diameter 4" 1051: Start downhole vertical logging of 251-2 1205: Complete vertical profile. Pull up transducer & decon 1210: Set up for slug test. Set up Troll 200 to log 1 sec. intervals. 1251: Set Troll 200 @ 20.72' below water level 1335: Deploy additional Druck transducer. 1351: Insert slug 1354: Pull slug out 1356: Insert slug 1358: Pull slug out 1402: Insert slug 1404: Pull slug out. Clean up. 1450: Depart site.			
VISITORS ON SITE: NA		CHANGES FROM PLANS AND SPECIFICATIONS, AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS: NA	
WEATHER sunny / hot : low 72°F, high 90°F		IMPORTANT TELEPHONE CALLS: None NA	
APTIM PERSONNEL ON SITE: Simone Boudreau			
SIGNATURE:		DATE: 7-26-22	



FIELD ACTIVITY DAILY LOG

Daily Log	Date:	7	27	22
	No.			
	Sheet	1	Of	1

PROJECT NAME: Stringfellow PAT Project Project Number: 500520

FIELD ACTIVITY SUBJECT: Pump test on 251A-2; Downhole vertical profiling (post) Task number:

DESCRIPTION OF DAILY ACTIVITIES AND EVENTS,

1300: Arrive on site. Set up at well 251A-3. Greg Smith on site.

1310: FC 251A-3: DTW = 55.69' btoe
TD = 176' btoe 4" inside diameter

Will need longer rugged cable. We have 3 100' cables that do not connect. Call In-Situ to figure out cable situation. There is no way to connect 2 rugged cables so we will need to order a separate, longer cable.

1350: Greg Smith finds a 200' cable in the SD office. Will have it shipped overnight

1400: Phil Harte on site. Set up on well 251A-2 for pumping.

1602: Begin pumping. Pump intake 1.25' off well bottom.

1700: DTW = 56.6' btoe

~1810: Stop pumping. Clean up.

1900: Depart site.

VISITORS ON SITE:

NA

CHANGES FROM PLANS AND SPECIFICATIONS, AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS:

NA

WEATHER

sunny / hot high 92°F

IMPORTANT TELEPHONE CALLS:

None NA

APTIM PERSONNEL ON SITE: Simone Boudreau

SIGNATURE:

DATE: 7-27-22



FIELD ACTIVITY DAILY LOG

Daily Log	Date:	7	28	22
	No.			
	Sheet	1	Of	1

PROJECT NAME: Stringfellow PAT Project	Project Number: S00S20
FIELD ACTIVITY SUBJECT: Pre Downhole vertical Profile of FC 251A-3	Task number:

DESCRIPTION OF DAILY ACTIVITIES AND EVENTS,

1200: Receive call from Phil Harte (USGS) that 200' cable for Troll 200 data logger arrived. Move to site

1330: Arrive on site. Phil & Greg on site. Injected 170 gal into FC 251A-2 in the morning.

1334: FC 251A-3 DTW = 55.68' btoe
TD = 174.60

1344: Start vertical profile log.

1720: (Pre) vertical profile complete. Clean up.

1800: Depart site

VISITORS ON SITE:

NA

CHANGES FROM PLANS AND SPECIFICATIONS, AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS:

NA

WEATHER

Sunny, hot high 93°F

IMPORTANT TELEPHONE CALLS:

None

NA

APTIM PERSONNEL ON SITE: Simone Boudreau

SIGNATURE:

DATE: 7-28-22



FIELD ACTIVITY DAILY LOG

Daily Log	Date:	7	29	22
	No.			
	Sheet	1	Of	2

PROJECT NAME: Stringfellow PAT Project		Project Number: 500520	
FIELD ACTIVITY SUBJECT: Slug tests ; Pumping ; Post vertical Profile of 251A-3		Task number:	
DESCRIPTION OF DAILY ACTIVITIES AND EVENTS,			
0800: Arrive on site. Set up on well FC 251A-3.			
0855: Gauge well. DTW = 55.71' btoe			
0900: Measure out braided propylene rope for slug tests. Set up Troll 200			
0940: Deploy Level Troll 200 to 19.80' in water column.			
1002: Deploy slug			
1008 Pull out slug			
<u>CR-10</u>			
1013: Deploy slug			
1019: Pull out slug		10:24	35.56
1028: Deploy slug		10:28	55.31
1035: Pull out slug		10:40	55.10
1040: Deploy slug		10:51	54.85
1046: Pull out slug			
1050: Deploy slug			
1056: Pull out slug.			
1100: Decon slug. Prepare for pumping			
1130: Set Geotech electric submersible pump 1.25' off bottom of well			
1217: Begin pumping			
Set up Troll 200 to log every 1 min. Lower Troll by ~1m (3.5 ft) every 3 minutes in water column above screen interval. Lower Troll by ~0.5 m (1.7 ft) every 3 minutes within screen interval. Screen 56-101'			
VISITORS ON SITE:		CHANGES FROM PLANS AND SPECIFICATIONS, AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS:	
N/A		NA	
WEATHER		IMPORTANT TELEPHONE CALLS:	
Sunny/hot low 72°F high 93°F		None	
APTIM PERSONNEL ON SITE: Simone Boudreau			
SIGNATURE:		DATE: 7-29-22	



FIELD ACTIVITY DAILY LOG

Daily Log	Date:	7	29	22
	No.			
	Sheet	2	Of	2

PROJECT NAME: Stringfellow PAT Project Project Number: 500520

FIELD ACTIVITY SUBJECT: Slug tests; Pumping; Post vertical Profile of 251A-3 Task number:

DESCRIPTION OF DAILY ACTIVITIES AND EVENTS,

1340: DTW = 56.35' btoe. Start log. 1 minute log intervals.

1639: Complete vertical profile. Stop log. Decon Level Troll.

1650: Depart site.

VISITORS ON SITE:

NA

CHANGES FROM PLANS AND SPECIFICATIONS, AND OTHER
SPECIAL ORDERS AND IMPORTANT DECISIONS:

NA

WEATHER

sunny/hot low 72°F
high 93°F

IMPORTANT TELEPHONE CALLS:

None

APTIM PERSONNEL ON SITE: Simone Boudreau

SIGNATURE:

DATE: 7-29-22



FIELD ACTIVITY DAILY LOG

Daily Log	Date	8	1	22
	No.			
	Sheet	1	Of	2

PROJECT NAME: Stringfellow PAT Project		Project Number: S00520	
FIELD ACTIVITY SUBJECT: Slug tests, Pre & Post vertical Profiling of 1011A		Task number:	
DESCRIPTION OF DAILY ACTIVITIES AND EVENTS,			
0800: Arrive at Agate st. well cluster. Call to Phil Harte			
0815: Call to Piotr to see if he has heard from USGS. They are filling up tank at DTSC facility.			
0830: USGS arrive. Begin setting up traffic control			
0939: Gauge FC 1011A. DTW = 17.43' btoe TD = 61.3' 3.75" inner diameter. Screen 53-87 ⁵⁵ 15-60'			
0944: Begin vertical profile log. Move Troll 200' 1m (~3.35') lower every 3 min. within water column above screened interval. Lower Troll 200 0.5 m (~1.7') every 3 min. when within screen interval.			
1056: Complete vertical profile. Prepare for slug tests			
1153: DTW = 17.42' btoe Set. Troll 200 16.37' below water level			
1156: Deploy slug			
1200: Pull out slug			
1203: Deploy slug			
1204: Pull out slug			
1206: Deploy slug) Slug Test 1			
1207: Pull out slug			
1209: Deploy slug) Slug Test 2			
1211: Pull out slug			
1213: Deploy slug) Slug Test 3			
1214: Pull out slug			
VISITORS ON SITE: NA		CHANGES FROM PLANS AND SPECIFICATIONS, AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS: NA	
WEATHER morning: sunny high 77°F afternoon: " high 95°F		IMPORTANT TELEPHONE CALLS: None	
APTIM PERSONNEL ON SITE: Simone Boudreau			
SIGNATURE:		DATE: 8-1-2022	



FIELD ACTIVITY DAILY LOG

Daily Log	Date:	8	1	22
	No.			
	Sheet	2	Of	2

PROJECT NAME: Stringfellow PAT Project	Project Number: 500520
FIELD ACTIVITY SUBJECT: Slug tests, Pre & Post Vertical Profile of 1011A	Task number:

DESCRIPTION OF DAILY ACTIVITIES AND EVENTS,

1220: Set up for pumping

1327: start logging for post vertical profile. DTW = 17.58' btoe

1444: Complete vertical profile. Disconnect & decon Troll 200.

Clean up site.

1600: Depart site

DTW @ 1348 = 17.59' btoe

VISITORS ON SITE:

NA

CHANGES FROM PLANS AND SPECIFICATIONS, AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS:

NA

WEATHER

morning: sunny high 77°F
afternoon: " 95°F

IMPORTANT TELEPHONE CALLS:

None

APTIM PERSONNEL ON SITE: Simone Goudreau

SIGNATURE:

DATE: 8-1-22

Project Name: Stringfellow PAT Project Project No. 500520
 Field Activity Subject: Aquifer Testing Activities
 Description of Daily Activities and Events: Slug Tests, Push/Pull on 1011B1, Nat. Gradient Test on 1029

0800: Arrive at DTSC facility. Greg (USGS) on site. Fill up 100 gal tank with water & salt. Move to Agate St. well cluster.
 0830: Set up work site: canopies around well FC 1011B1.
 0855: DTW = 17.39' TD = 134.78' ID of casing 3.75"
 0910: Tom Perina on site. Measure out rope for slug test.
 0920: Deploy data logger at 14.70' for slug tests.
 0933: Deploy slug slug Test 1
 0941: Pull out slug (14.81)
 0948: Drop in slug 14.77
 0957: Pull out slug 14.81 slug Test 2
 1008: Drop in slug 14.79
 1018: Pull out slug 14. slug Test 3
 Assess slug to check for leaking. Some water inside slug. Filled with clay soil.
 1025: Pull out Aqua Troll 200 & decan cable. Deploy transducer in adj. well (FC 1011B2).
 1053: Inject salt water into well FC 1011B1. Conductivity increasing in adjacent well (bedrock fractures)
 [70 mS/cm salt water 193 gal]
 1211: All water injected into well. Shut off
 1419: Start pumping.
 1428: DTW = 20.82' btop
 1445: T. Perina & J. Boudreau move to well FC 1029 for natural gradient test.
 1505: Start log with Troll 200 in 100 gal poly tank temp. 35.67°C
 1520: Take data logger out of tank & stick it in cold water bottle.
 1522: Deploy transducer in well. Coil up extra cable & stick it in casing well. Wait for temp. to stabilize before gravity-draining tank.
 1548: Open hose to drain tank.
 Inside diameter 5.7"

VISITORS ON SITE:

NA

CHANGES FROM PLANS AND SPECIFICATIONS AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS:
WEATHER CONDITIONS:

Sunny/clear; very hot
 morning 78°F; afternoon

IMPORTANT TELEPHONE CALLS:

NA

CB&I PERSONNEL ON SITE: S. Boudreau

SIGNATURE: _____

DATE: 8-3-2022

1635: Stop draining. Disconnect hose & tank.

1640: Disconnect troll comm & plug in large desiccant to leave in well

1724: Shut off pump in well 1011B1

1830: Depart site



Project Name: Stringfellow Project No. 500520
 Field Activity Subject: Salt Spike ; Push/Pull on well 251A-2; Natural Gradient
 Description of Daily Activities and Events: on well 1011A

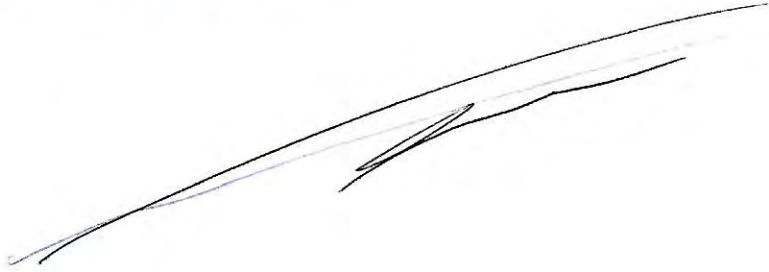
0800: Arrive at DTSC facility. Fill up water tank with water ; 10 lbs salt
 0825: Move to FC 251A-2. Set up equipment.
 0900: Open up extraction well vault. Will take depth readings off of digital meter inside vault. Open up CTN OW1 ; deploy
 0950: Deploy transducer (Troll 200) in well. Screen interval 47.3 to 77.3 transducer set at 17.42' in water column.
 1000: Gauge well with water level meter. DTW = 55.45' btoe (17.42')
 1025: FC 251A-2 DTW = 55.50'
 1100: Deploy pump to bottom of FC 251-2. Transducer in well @ 60' btoe. [Extraction well CTN-TW1 was shut off at 1000 on 8-3].
 1106: Turn on pump. Take photos of depth in TW1 since no transducer in that well.
 1158: Initial specific conductivity in CTN OW1 = 1206.6 $\mu S/cm$
 1200: Deploy salt slug into FC 251A-2. Wait for breakthrough time.
 1237: Reached breakthrough by this time (before this time)
 1331: Shut off pump.
 1345: Begin "Push" test
 1404: shut off pump.
 1425: T. Perina on site. Pull up Level Troll 200 ; decon it. Measure ; mark 22.5' on cable
 Move to well 1011A on Agate st.
 Set up traffic control. Set up for natural gradient test.
 1508: DTW = 17.47' btoe.
 1509: Deploy Troll 200 in 100 gal tank.
 1510: Start log. Temp. of tank ~ 33.86 - 34.87. Conductivity ~ 20,000 $\mu S/cm$
 1517: Move data logger to add water bottle to cool it down.
 1520: Deploy data logger in well 1011A.

VISITORS ON SITE: NA	CHANGES FROM PLANS AND SPECIFICATIONS AND OTHER SPECIAL ORDERS AND IMPORTANT DECISIONS: NA
WEATHER CONDITIONS: sunny, hot ^(overcast) morning 75°F afternoon 92°F	IMPORTANT TELEPHONE CALLS: NA
CB&I PERSONNEL ON SITE: <u>S. Boddreau / T. Perina</u>	
SIGNATURE: 	DATE: <u>8-4-2022</u>

1530: Begin draining tank

1610: Tank emptied: Clean up. Move back to FC 2S1A-2 to return traffic control equipment.

1645: Depart site



Appendix B

Recorded Data

(electronic files)

Appendix C

Hydraulic Analysis

[example output in spreadsheets]

Appendix D

Transport Analysis

[example output in spreadsheets]